

Score Contributor 1: Selection Sensibility Score (SSS)

BASE PROMPT:

This is how I had structured the prompt while running with Gemma 4. Please use it for Qwen 235 B

```
"""
selection_sensibility/common.py
"""
import re

BAND_LABELS = {
    4: "Fully Sensible",
    3: "Minor Deviation",
    2: "Moderate Violation",
    1: "Major Violation",
    0: "Severe Violation",
}

STYLE_ADAPTERS = {
    "Progressive Reveal": (
        """*** STYLE ADAPTER: PROGRESSIVE REVEAL ***\n"""
        "In Progressive Reveal, elements are cumulatively drawn onto the canvas.\n"
        "- 'Newly Revealed Elements': For this evaluation, identify the specific diagram elements "
        "that were just drawn or rendered in the CURRENT frame that were not present in the PREVIOUS frame."
    ),
    "Alpha Masking": (
        """*** STYLE ADAPTER: ALPHA MASKING ***\n"""
```

"In Alpha Masking, the entire diagram is visible, but active elements are fully opaque "

"while inactive elements are masked (transparent or dimmed).\n"

"- 'Newly Revealed Elements': For this evaluation, identify the specific diagram elements "

"that transitioned to FULL OPACITY in the CURRENT frame. (Do not evaluate elements that faded "

"back to a dimmed state; focus only on what is actively highlighted now)."

),

"Colour Pop": (

*** STYLE ADAPTER: COLOUR POP ***\n"

"In Colour Pop, the entire diagram starts in greyscale, and active elements permanently "

"transition to full color.\n"

"- 'Newly Revealed Elements': For this evaluation, identify the specific diagram elements "

"that transitioned from greyscale to FULL COLOR in the CURRENT frame."

),

"Hopping Bounding Box": (

*** STYLE ADAPTER: HOPPING BOUNDING BOX ***\n"

"In Hopping Bounding Box, the diagram itself does not move or change. Instead, a discrete "

"highlight box jumps to enclose the active region.\n"

"- 'Newly Revealed Elements': For this evaluation, identify the specific underlying diagram "

"elements (nodes, blocks, etc. from the XML) that are ENCLOSED by the new position of the "

"bounding box in the CURRENT frame. You are scoring the selection of the elements inside "

"the box, not the box itself."

),

"Sliding Bounding Box": (

*** STYLE ADAPTER: SLIDING BOUNDING BOX ***\n"

```

        "In Sliding Bounding Box, the diagram itself does not
        move or change. Instead, a highlight "
        "box smoothly slides to enclose the active region.\n"
        "- 'Newly Revealed Elements': For this evaluation, id
        entify the specific underlying diagram "
        "elements (nodes, blocks, etc. from the XML) that are
        ENCLOSED by the bounding box once it "
        "has come to rest in the CURRENT frame. You are scori
        ng the selection of the elements inside "
        "the box, not the box itself."
    )
}

```

RUBRIC_TEXT = ""

You will score ONE combined severity band (0-4) per timestep, but you must reason about the two underlying criteria SEPARATELY before collapsing them into a single band. Do not skip straight to a number.

--- CRITERION A: ELEMENT APPROPRIATENESS ---

Given the spatial layout, the underlying flow/hierarchy of the ORIGINAL

diagram, and what has already been targeted/revealed in prior timesteps, judge whether

the newly targeted element or group of elements at THIS timestep is the right thing to focus on now.

- It is correct if it logically follows from what was already shown, OR if

- this is the first reveal and it is a sensible entrypoint in to the diagram

- (per the XML hierarchy/edges).

- A MISSING significant element (one that logically belongs in this reveal

- group and is not yet present) is a heavier violation than a

n EXTRA element

that could have been deferred to a later step.

- This criterion is also the catch-all for cases where the animation STYLE is

technically correct but is being applied to the wrong element at the wrong time.

--- CRITERION B: GROUP COHERENCE ---

Judge whether the elements targeted TOGETHER at this timestep are related

enough to justify being grouped. A valid group reflects at least one

of:

(i) structural relation (e.g. parallel elements, or unconnected elements

that are spatially/structurally close per the XML ``block`` /

``composite_whole`` nesting),

(ii) logical relation (components of the same sub-structure, frames of the

same picture),

(iii) semantic relation (elements representing the same idea).

- If two elements are grouped in the SAME timestep but the XML ``edge`` data

shows a non-trivial PREREQUISITE dependency between them (one must

logically/visually exist before the other for the reveal to make sense),

that is a violation – trivial/instantaneous dependencies (e.g. a node and

its own text label) do not count.

- Grouping elements that belong to unrelated sub-diagrams/branches with no

structural, logical, or semantic link is penalized.

--- COMBINING A AND B INTO ONE BAND ---

4 - Fully Sensible: Correct element/group choice per Criterion A AND a

coherent, justified group per Criterion B. No missing elements, no

dependency violations.

3 - Minor Deviation: Core choice is correct, but there is one small avoidable

imperfection (e.g. one extra/deferrable element included, or the group

could have been split slightly more cleanly) – nothing significant is

missing and no dependency is violated.

2 - Moderate Violation: EITHER a non-critical element is missing from what

should logically be present at this step, OR the group mixes in one

loosely-related element without violating a hard dependency – but the

overall intent of the step is still recognizable.

1 - Major Violation: EITHER a significant/necessary element is missing such

that the reveal does not make sense on its own, OR elements are grouped

despite a non-trivial prerequisite dependency between them, OR the group

mixes clearly unrelated structural/semantic elements.

0 - Severe Violation: The element/group choice is essentially arbitrary or

incoherent relative to the diagram – wrong entrypoint, elements from

unrelated parts of the diagram merged with no justification, or the step

actively contradicts the diagram's flow/hierarchy.

Tie-break rule: if Criterion A and Criterion B would independently point to different bands, report the LOWER (worse) band as your combined band, and say explicitly in the rationale which criterion drove the score. If BOTH criteria are independently moderate-or-worse (band ≤ 2), go one band lower than the worse of the two (violations compound), down to a floor of 0.

```
"".strip()
```

```
OUTPUT_FORMAT_INSTRUCTIONS = ""
```

Respond in EXACTLY this structure, with these literal section headers:

```
### RATIONALE
```

<Write a thorough step-by-step rationale. First identify what is NEWLY TARGETED in the current frame relative to the previous frame according to the Style Adapter, mapping it to specific XML IDs. Then analyze Criterion A (element appropriateness) referencing specific XML ids and what was already targeted. Then analyze Criterion B (group coherence) referencing specific XML ids and any relevant edges. Then explain how you combined them into the final band using the tie-break rule.>

```
### CRITERION_A_BAND: <integer 0-4, your standalone judgment for Criterion A only>
```

```
### CRITERION_B_BAND: <integer 0-4, your standalone judgment for Criterion B only>
```

```
### FINAL_BAND: <integer 0-4, the combined band per the tie-break rule>
```

```
### LABEL: <one of: Fully Sensible | Minor Deviation | Moderate Violation | Major Violation | Severe Violation>
```

Do not output anything after the LABEL line.

```

"".strip()

def build_system_instruction():
    return (
        "You are a meticulous evaluator of diagram animations. You are an "
        "expert at reading hierarchical XML diagram schemas (blocks, standalone "
        "nodes, child nodes, raster nodes, composite wholes/parts, and edges) and "
        "mapping them onto what is visually targeted/revealed at each animation timestep. "
        "You are strict, consistent, and skeptical of animations that merely "
        "'look fine' but do not respect the diagram's actual structure and "
        "dependencies. You always ground your reasoning in specific element ids "
        "from the provided XML, never vague visual impressions alone.\n\n"
        + RUBRIC_TEXT
    )

def build_user_prompt(xml_text, timestep_idx, total_steps, prior_summary, style_name="Progressive Reveal"):
    # Fetch the corresponding adapter text, default to Progressive Reveal if not found
    style_adapter_text = STYLE_ADAPTERS.get(style_name, STYLE_ADAPTERS["Progressive Reveal"])

    return (
        f"This is timestep {timestep_idx} of {total_steps} in a '{style_name}' "
        "animation of the diagram below.\n\n"
        "You are given, in order:\n"
        "1. The ORIGINAL fully-revealed diagram image (ground

```

```

truth layout).\n"
    "2. The PREVIOUS timestep's rendered frame (state before this reveal). "
    "Omitted if this is timestep 1.\n"
    "3. The CURRENT timestep's rendered frame (state after this reveal).\n\n"
    f"--- CUMULATIVE ELEMENTS ALREADY TARGETED BEFORE THIS STEP ---\n"
    f"{prior_summary}\n\n"
    "--- FULL XML DIAGRAM SCHEMA ---\n"
    f"{xml_text}\n\n"
    f"{style_adapter_text}\n\n"
    "Identify what is NEW in the CURRENT frame relative to the PREVIOUS "
    "frame (or, if timestep 1, what the entrypoint reveal consists of) based strictly "
    "on the Style Adapter definition above, map it onto ids in the XML, and score it per the rubric.\n\n"
    + OUTPUT_FORMAT_INSTRUCTIONS
)

_BAND_RE = {
    "criterion_a_band": re.compile(r"###\s*CRITERION_A_BAND:\s*([0-4])", re.IGNORECASE),
    "criterion_b_band": re.compile(r"###\s*CRITERION_B_BAND:\s*([0-4])", re.IGNORECASE),
    "final_band": re.compile(r"###\s*FINAL_BAND:\s*([0-4])", re.IGNORECASE),
    "label": re.compile(r"###\s*LABEL:\s*(.+)", re.IGNORECASE),
}

_RATIONALE_RE = re.compile(
    r"###\s*RATIONALE\s*(.*?)(?=###\s*CRITERION_A_BAND)", re.IGNORECASE | re.DOTALL
)

```



```

def parse_judge_output(raw_text):
    result = {
        "rationale": None,
        "criterion_a_band": None,
        "criterion_b_band": None,
        "final_band": None,
        "label": None,
        "is_valid": False,
        "raw_output": raw_text,
    }

    rationale_match = _RATIONALE_RE.search(raw_text)
    if rationale_match:
        result["rationale"] = rationale_match.group(1).strip()

    for key, pattern in _BAND_RE.items():
        m = pattern.search(raw_text)
        if m:
            if key == "label":
                result[key] = m.group(1).strip().splitlines()[0].strip()
            else:
                result[key] = int(m.group(1))

    if result["final_band"] is not None:
        result["is_valid"] = True
        if not result["label"]:
            result["label"] = BAND_LABELS.get(result["final_band"] and "")

    return result

def read_file_safely(filepath):
    import os
    if os.path.exists(filepath):

```

```

        with open(filepath, "r", encoding="utf-8", errors="replace") as f:
            return f.read().strip()
    return ""

_IMAGE_EXTS = (".png", ".jpg", ".jpeg")
_FRAMES_DIR_CANDIDATES = ("frames", "animation_frames", "video_frames", "anim_frames")

def locate_sample_files(sample_dir):
    import os
    if not os.path.isdir(sample_dir):
        return None, None, None, f"sample_dir does not exist: {sample_dir}"

    entries = os.listdir(sample_dir)
    top_files = [e for e in entries if os.path.isfile(os.path.join(sample_dir, e))]
    top_dirs = [e for e in entries if os.path.isdir(os.path.join(sample_dir, e))]

    image_files = [f for f in top_files if f.lower().endswith(_IMAGE_EXTS)]
    original_img = None
    for f in image_files:
        low = f.lower()
        if low.startswith(("original", "gt", "ground_truth", "source")):
            original_img = os.path.join(sample_dir, f)
            break
    if original_img is None and len(image_files) == 1:
        original_img = os.path.join(sample_dir, image_files[0])

    xml_files = [f for f in top_files if f.lower().endswith(".xml")]

```

```

    xml_path = os.path.join(sample_dir, xml_files[0]) if len
(xml_files) >= 1 else None
    if len(xml_files) > 1:
        preferred = [f for f in xml_files if "diagram" in f.lower() or "schema" in f.lower()]
        if preferred:
            xml_path = os.path.join(sample_dir, preferred[0])

frames_dir = None
for cand in _FRAMES_DIR_CANDIDATES:
    for d in top_dirs:
        if d.lower() == cand:
            frames_dir = os.path.join(sample_dir, d)
            break
    if frames_dir:
        break
if frames_dir is None and len(top_dirs) == 1:
    frames_dir = os.path.join(sample_dir, top_dirs[0])

missing = []
if original_img is None:
    missing.append(f"original image (found top-level images: {image_files or 'none'})")
if xml_path is None:
    missing.append(f"xml schema file (found top-level .xml files: {xml_files or 'none'})")
if frames_dir is None:
    missing.append(f"frames subdirectory (found subdirectories: {top_dirs or 'none'})")

if missing:
    return original_img, xml_path, frames_dir, "missing: " + "; ".join(missing)

return original_img, xml_path, frames_dir, None

```

Results for Alpha Masking

attachment:e4fc1132-5da2-4638-ab41-2ad8db558d78:CVPR_2025_pipe00004_video_tikz_A
M.mp4

```
{
  "sample_dir": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ",
  "model": "google/gemma-4-31B-it",
  "n_timesteps": 15,
  "timesteps": [
    {
      "rationale": "**Criterion A: Element Appropriateness**\nThis is the first timestep (entrypoint). The animation reveals the initial input data of the pipeline: the multi-view images (`I_stack_1`, `I_stack_2`, `I_stack_3`) and their corresponding segmentation masks (`M_stack_1`, `M_stack_2`, `M_stack_3`), along with their labels (`I_1_K_lbl`, `M_1_K_lbl`) and the general \"Views\" label (`views_lbl`). This is a highly sensible entrypoint as it establishes the inputs before showing the processing steps. The inclusion of the bottom caption text is also appropriate for setting the context.\n\n**Criterion B: Group Coherence**\nThe elements rev
```

ealed together are logically and structurally linked. The image stacks and mask stacks are parallel inputs in the diagram's flow. They are spatially adjacent and semantically related (images and their masks). There are no prerequisite dependency violations here, as these are the starting nodes of the graph. The grouping is coherent and reflects the "Given..." part of the caption.

Combined Band

Both Criterion A and Criterion B are fully satisfied. The reveal is logical, complete for the intended step, and coherent."

```
        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n**Criterion A: Element Appropriateness**\nThis is the first timestep (entrypoint). The animation reveals the initial input data of the pipeline: th
```

e multi-view images (`I\_stack\_1`, `I\_stack\_2`, `I\_stack\_3`) and their corresponding segmentation masks (`M\_stack\_1`, `M\_stack\_2`, `M\_stack\_3`), along with their labels (`I\_1\_K\_lbl`, `M\_1\_K\_lbl`) and the general "Views" label (`views\_lbl`). This is a highly sensible entrypoint as it establishes the inputs before showing the processing steps. The inclusion of the bottom caption text is also appropriate for setting the context.

Criterion B: Group Coherence

The elements revealed together are logically and structurally linked. The image stacks and mask stacks are parallel inputs in the diagram's flow. They are spatially adjacent and semantically related (images and their masks). There are no prerequisite dependency violations here, as these are the starting nodes of the graph. The grouping is coherent and reflects the "Given..." part of the caption.

Combined Band

Both Criterion A and Criterion B are fully satis

fied. The reveal is logical, complete for the intended step, and coherent.\n\n### CRITERION_A_BAND: 4\n## CRITERION_B_BAND: 4\n## FINAL_BAND: 4\n### LABEL: Fully Sensible",
 "timestep": 1,
 "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/1_CVPR_2025_pipe00004_frames_tikz_AM.png"
 },
 {
 "rationale": "**Criterion A: Element Appropriateness**\n\nThe previous timestep established the input views and masks (`I\_stack\_1`, `M\_stack\_1`) and the general context via the caption. The current timestep reveals the 3D Gaussian Splatting model (`G\_box` and its image `G\_box\_img`) and the edge connecting the input images to it (`arrow\_I\_G`). This is the logically correct next step in the pipeline: after providing the input views, the animation shows the model being pretrained on those views, as explicitly stated in the caption (\n"a 3D Gau

ssian Splatting model is first pretrained on the input views"). The reveal is perfectly aligned with the diagram's flow and the accompanying text.

Criterion B: Group Coherence

The elements revealed together are the model block (``G_box``) and the directional edge (``arrow_I_G``) that leads into it from the already-revealed input images. This is a highly coherent group as it represents a single logical operation: the flow of data from inputs to the model. There are no dependency violations; the source of the arrow (``I_stack_1``) was already present, and the target (``G_box``) is revealed simultaneously.

Combined Band

Both Criterion A and Criterion B are fully satisfied. The reveal is logically sequenced, structurally sound, and matches the semantic intent of the diagram and caption.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,

```



```

        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n**Criterion A: Element Appropriateness**\n\nThe previous timestep established the input views and masks (`I_stack_1`, `M_stack_1`) and the general context via the caption. The current timestep reveals the 3D Gaussian Splatting model (`G_box` and its image `G_box_img`) and the edge connecting the input images to it (`arrow_I_G`). This is the logically correct next step in the pipeline: after providing the input views, the animation shows the model being pretrained on those views, as explicitly stated in the caption (`"a 3D Gaussian Splatting model is first pretrained on the input views"`). The reveal is perfectly aligned with the diagram's flow and the accompanying text.\n\n**Criterion B: Group Coherence**\n\nThe elements revealed together are the model block (`G_box`) and the directional edge (`arrow_I_G`) that lead

```

s into it from the already-revealed input images. This is a highly coherent group as it represents a single logical operation: the flow of data from inputs to the model. There are no dependency violations; the source of the arrow (`I\_stack\_1`) was already present, and the target (`G\_box`) is revealed simultaneously.

Combined Band

Both Criterion A and Criterion B are fully satisfied. The reveal is logically sequenced, structurally sound, and matches the semantic intent of the diagram and caption.

CRITERION_A_BAND: 4
 ### CRITERION_B_BAND: 4
 ### FINAL_BAND: 4
 ### LABEL: Fully Sensible",

 "timestep": 2,

 "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/2_CVPR_2025_pipe00004_frames_tikz_AM.png"

 },

 {

 "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n- In the previous timestep

p, the animation revealed the input views (`I\_stack\_1`), the 3DGS model (`G\_box`), and the caption stating the model is pretrained.
 - In the current timestep, the newly revealed elements are the edge `arrow\_G\_D` (`"Render Depth`"), the depth map stack `D\_stack\_1` (including `D\_stack\_1\_img`), and the label `D\_1\_K\_lbl`.
 - According to the XML, `arrow\_G\_D` connects the previously revealed `G\_box` to `D\_stack\_1`. This is the logical next step in the pipeline: after the model is pretrained, it is used to render depth maps. The reveal perfectly matches the caption `"Depth maps are then rendered... from this model."
 - Criterion A is fully satisfied.
 2. **Analyze Criterion B (Group Coherence):**
 - The elements revealed together are: the arrow (`arrow\_G\_D`), the target node (`D\_stack\_1`), and its associated label (`D\_1\_K\_lbl`).
 - These elements form a single logical unit (an action and its result). There

are no prerequisite dependency violations; the source of the arrow (`G\_box`) was already present.\n - Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n - Both Criterion A and Criterion B are scored as 4. Following the tie-break rule, the final band is 4.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n - In the previous timestep, the animation revealed the input views (`I_stack_1`), the 3D GS model (`G_box`), and the caption stating the model is pretrained.\n - In the current timestep, the newly revealed elements are the edge `arrow_G_D` (`Render Depth`), the depth map stack `D_stack_1` (including `D_stack_1_img`), and the label `D_1_K_1`

```

```

bl`.\\n    - According to the XML, `arrow_G_D` connects the previously revealed `G_box` to `D_stack_1`. This is the logical next step in the pipeline: after the model is pretrained, it is used to render depth maps. The reveal perfectly matches the caption \"Depth maps are then rendered... from this model.\\n    - Criterion A is fully satisfied.\\n\\n2. **Analyze Criterion B (Group Coherence):**\\n    - The elements revealed together are: the arrow (`arrow_G_D`), the target node (`D_stack_1`), and its associated label (`D_1_K_lbl`).\\n    - These elements form a single logical unit (an action and its result). There are no prerequisite dependency violations; the source of the arrow (`G_box`) was already present.\\n    - Criterion B is fully satisfied.\\n\\n3. **Combining A and B:**\\n    - Both Criterion A and Criterion B are scored as 4. Following the tie-break rule, the final band is 4.\\n\\n### CRITERION_A_BAND: 4\\n### CRITERION_B_BAND: 4

```

```

\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",
    "timestep": 3,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/3_CVPR_2025_pipe00004_frames_tikz_AM.png"
},
{
    "rationale": "**Criterion A: Element Appropriateness**\n\nThe previous timestep focused on the \"Gaussians with Bear Statue\" section, specifically rendering depth maps (`D_stack_1`). The current timestep transitions to the second major section of the diagram: \"Inferring Depth-Guided Inpainting Mask\". The newly revealed elements are the contents of `box_I2_M2` (specifically `I2_img`, `M2_img`, and their labels `I2_lbl`, `M2_lbl`) and `box_I1_M1` (specifically `I1_img`, `M1_img`, and their labels `I1_lbl`, `M1_lbl`). This is a logical progression as the process moves from the initial 3D model to the processing of individual view images and masks to infer the inpaint

```

ing mask. The caption explicitly states \"individual view images and masks are processed,\" which perfectly matches the revealed elements.

Criterion B: Group Coherence

The elements revealed together are two parallel blocks: ``box_I2_M2`` and ``box_I1_M1``. These are structurally identical (each containing an image and a mask for a specific view) and semantically related as they represent the \"individual view images and masks\" mentioned in the caption. There are no prerequisite dependencies between these two blocks that would require one to be shown before the other; they are parallel inputs to the subsequent pipeline. The grouping is highly coherent.

Combined Band

Both Criterion A and Criterion B are fully satisfied. The transition is logical, the elements match the caption, and the grouping reflects the structural symmetry of the diagram.

\"criterion_a_band\":

4,

```

        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sens
ible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n**Criterion A: E
lement Appropriateness*\n
The previous timestep focu
sed on the \"Gaussians wit
h Bear Statue\" section, s
pecifically rendering dept
h maps (`D_stack_1`). The
current timestep transitio
ns to the second major sec
tion of the diagram: \"Inf
erring Depth-Guided Inpain
ting Mask\". The newly rev
ealed elements are the con
tents of `box_I2_M2` (spec
ifically `I2_img`, `M2_img
`, and their labels `I2_lb
l`, `M2_lbl`) and `box_I1_
M1` (specifically `I1_img
`, `M1_img`, and their lab
els `I1_lbl`, `M1_lbl`). T
his is a logical progressi
on as the process moves fr
om the initial 3D model to
the processing of individu
al view images and masks t
o infer the inpainting mas
k. The caption explicitly
states \"individual view i
mages and masks are proces

```



```
sed,\" which perfectly matches the revealed elements.\n\n**Criterion B: Group Coherence**\n\nThe elements revealed together are two parallel blocks: `box_I2_M2` and `box_I1_M1`. These are structurally identical (each containing an image and a mask for a specific view) and semantically related as they represent the \"individual view images and masks\" mentioned in the caption. There are no prerequisite dependencies between these two blocks that would require one to be shown before the other; they are parallel inputs to the subsequent pipeline. The grouping is highly coherent.\n\n**Combined Band**\n\nBoth Criterion A and Criterion B are fully satisfied. The transition is logical, the elements match the caption, and the grouping reflects the structural symmetry of the diagram.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible\",  
    \"timestep\": 4,  
    \"frame path\": \"/arch
```

```
ive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/4_CVPR_2025_pipe00004_frames_tikz_AM.png"
```

```
},  
{
```

```
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n    - In the previous timestep, the animation revealed the input images and masks within `box_I2_M2` and `box_I1_M1` (specifically `I2_img`, `M2_img`, `I1_img`, `M1_img`).\n    - In the current timestep, the newly revealed elements are the background-isolated images `I2B_img` and `I1B_img`, along with the edges `arrow_M2_I2B` and `arrow_M1_I1B`.\n    - According to the XML, `M2_img` is the source for `arrow_M2_I2B` which targets `I2B_img`, and `M1_img` is the source for `arrow_M1_I1B` which targets `I1B_img`.\n    - This is a logically correct progression: the masks revealed in the previous step are now being applied to the images to create the isolated background views. The caption \"The target object mas
```

k is applied to isolate the visible background region in each view\" perfectly matches this transition.\n

- Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n
 - The elements revealed together are `I2B\_img`, `I1B\_img`, and their corresponding incoming edges from the masks.\n
 - These elements are structurally and logically parallel: they represent the same operation (masking) applied to two different view-mask pairs.\n
 - There are no prerequisite dependencies between `I2B\_img` and `I1B\_img` themselves; they are independent parallel branches of the same process.\n
 - Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n
 - Both Criterion A and Criterion B are scored as 4. Following the tie-break rule, the final band is 4.",

```

      "criterion_a_band":
4,
      "criterion_b_band":
4,
      "final_band": 4,
      "label": "Fully Sens

```

```

ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - In the p
revious timestep, the anim
ation revealed the input i
mages and masks within `bo
x_I2_M2` and `box_I1_M1`
(specifically `I2_img`, `M
2_img`, `I1_img`, `M1_img
`).\n    - In the current t
imestep, the newly reveale
d elements are the backgro
und-isolated images `I2B_i
mg` and `I1B_img`, along w
ith the edges `arrow_M2_I2
B` and `arrow_M1_I1B`.\n
    - According to the XML, `M
2_img` is the source for `
arrow_M2_I2B` which target
s `I2B_img`, and `M1_img`
is the source for `arrow_M
1_I1B` which targets `I1B_
img`.\n    - This is a logi
cally correct progression:
the masks revealed in the
previous step are now bein
g applied to the images to
create the isolated backgr
ound views. The caption
`"The target object mask i
s applied to isolate the v
isible background region i
n each view`" perfectly ma

```

tches this transition.\n

- Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n
 - The elements revealed together are `I2B\_img`, `I1B\_img`, and their corresponding incoming edges from the masks.\n
 - These elements are structurally and logically parallel: they represent the same operation (masking) applied to two different view-mask pairs.\n
 - There are no prerequisite dependencies between `I2B\_img` and `I1B\_img` themselves; they are independent parallel branches of the same process.\n
 - Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n
 - Both Criterion A and Criterion B are scored as 4. Following the tie-break rule, the final band is 4.\n\n###
 - # CRITERION_A_BAND: 4\n###
 - CRITERION_B_BAND: 4\n###
 - FINAL_BAND: 4\n###
 - LABEL: Fully Sensible",
 - "timestep": 5,
 - "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/5_CVPR_2025_pipe00004_fram

```

es_tikz_AM.png"
    },
    {
        "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- The previous timestep fo
cused on the application o
f masks to isolate backgro
und regions (revealing `I2
B_img` and `I1B_img`).\n
- The current timestep rev
eals the depth map `D2_img
` and its associated label
`D2_lbl`.\n
- Looking at
the XML edges, there is a
direct dependency: `arrow_
I2B_D2` connects `I2B_img`
(already revealed) to `D2_
img`.\n
- The caption ex
plicitly states: \"Using r
endered depth maps, backgr
ound pixels from secondary
views are projected into 3
D space.\" This logically
justifies the reveal of th
e depth map `D2_img` as th
e next step in the pipelin
e.\n
- The reveal is app
ropriate and follows the s
tructural flow of the diag
ram.\n\n2. **Analyze Crite
rion B (Group Coherence):*
*\n
- The elements revea
led together are `D2_img`
and `D2_lbl`.\n
- These

```

are a node and its corresponding text label. Per the rubric, this is a trivial dependency and constitutes a perfectly coherent group.

- There are no unrelated elements mixed into this timestep.

3. **Combining A and B:**

- Criterion A is a 4 (Fully Sensible).

- Criterion B is a 4 (Fully Sensible).

- The combined band is 4.

```

{
  "criterion_a_band":
4,
  "criterion_b_band":
4,
  "final_band": 4,
  "label": "Fully Sensible",
  "is_valid": true,
  "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n  - The previous timestep focused on the application of masks to isolate background regions (revealing `I2B_img` and `I1B_img`).\n  - The current timestep reveals the depth map `D2_img` and its associated label `D2_lbl`.\n  - Looking at the XML edges, there is a direct

```

dependency: `arrow\_I2B\_D2` connects `I2B\_img` (already revealed) to `D2\_img`.\n

- The caption explicitly states: \"Using rendered depth maps, background pixels from secondary views are projected into 3D space.\" This logically justifies the reveal of the depth map `D2\_img` as the next step in the pipeline.\n
- The reveal is appropriate and follows the structural flow of the diagram.\n\n2. **Analyze Criterion B (Group Coherence):**\n
- The elements revealed together are `D2\_img` and `D2\_lbl`.\n
- These are a node and its corresponding text label. Per the rubric, this is a trivial dependency and constitutes a perfectly coherent group.\n
- There are no unrelated elements mixed into this timestep.\n\n3. **Combining A and B:**\n
- Criterion A is a 4 (Fully Sensible).\n
- Criterion B is a 4 (Fully Sensible).\n
- The combined band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Full


```

y Sensible",
    "timestep": 6,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/6_CVPR_2025_pipe00004_frames_tikz_AM.png"
},
{
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n
- In the previous timestep, the animation revealed the depth map `D2_img` and the `\"Project\"` edge `arrow_D2_I12B`.\n
- In the current timestep, the newly revealed elements are the raster node `I12B_img` (the image showing the projected background pixels) and its associated label `I12B_lbl`.\n
- According to the XML, `I12B_img` is the target of the edge `arrow_D2_I12B` (from `D2_img`) and `arrow_I1B_I12B` (from `I1B_img`). Since the projection process was initiated in the previous step, revealing the result of that projection (`I12B_img`) is the logically correct next step in the flow.\n
- Criterion A is fully sat

```

isfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n - The elements revealed together are `I12B\_img` and its label `I12B\_lbl`.\n - Per the rubric, a node and its own text label are considered a trivial dependency and constitute a coherent group.\n - There are no other unrelated elements mixed into this timestep.\n - Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n - Both Criterion A and Criterion B are scored as 4 (Fully Sensible). Following the tie-break rule, the final band is 4.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n - In the previous timestep, the animation revealed the depth map `D2_img` and the \"Project\" edge `arrow_D2_I12B

```

``.\n` - In the current timestep, the newly revealed elements are the raster node ``I12B_img`` (the image showing the projected background pixels) and its associated label ``I12B_lbl``.
``.\n` - According to the XML, ``I12B_img`` is the target of the edge ``arrow_D2_I12B`` (from ``D2_img``) and ``arrow_I1B_I12B`` (from ``I1B_img``). Since the projection process was initiated in the previous step, revealing the result of that projection (``I12B_img``) is the logically correct next step in the flow.
``.\n` - Criterion A is fully satisfied.
`\n2. **Analyze Criterion B (Group Coherence):**\n` - The elements revealed together are ``I12B_img`` and its label ``I12B_lbl``.
``.\n` - Per the rubric, a node and its own text label are considered a trivial dependency and constitute a coherent group.
``.\n` - There are no other unrelated elements mixed into this timestep.
``.\n` - Criterion B is fully satisfied.
`\n3. **Combining A and B:**\n` - Both Criterion A and Criteri

```

on B are scored as 4 (Fully Sensible). Following the tie-break rule, the final band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",
    "timestep": 7,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/7_CVPR_2025_pipe00004_frames_tikz_AM.png"
  },
  {
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n- In the previous timestep, the animation revealed the process of projecting background points onto the reference view, culminating in the image `I12B_img` (the combined background projection).\n- The current timestep reveals `Mprime1_img` (the refined inpainting mask).\n- According to the XML, there is a direct edge `arrow_I12B_Mprime1` from `I12B_img` to `Mprime1_img`.\n- The caption explicitly states that \"Excluding these visi

```

ble background pixels refines the original mask into a smaller inpainting mask, which perfectly describes the transition from the projected pixels (``I12B_img``) to the resulting mask (``Mprime1_img``).
 - This is the logically correct next step in the "Inferring Depth-Guided Inpainting Mask" sequence.
 2. **Analyze Criterion B (Group Coherence):**
 - Only one primary element is revealed in this step: ``Mprime1_img`` (and its associated label ``Mprime1_lbl`` which is visually tied to it).
 - Since it is a single logical unit following a direct dependency from the previous step, there are no grouping conflicts or unrelated elements mixed in. The group is perfectly coherent.
 3. **Combining A and B:**
 - Criterion A is 4 (Fully Sensible).
 - Criterion B is 4 (Fully Sensible).
 - The final band is 4.
 "criterion_a_band":
 4,
 "criterion_b_band":
 4,

```
"final_band": 4,  
"label": "Fully Sensible",  
"is_valid": true,  
"raw_output": "### R  
ATIONALE\n1. **Analyze Cri  
terion A (Element Appropri  
ateness):**\n    - In the p  
revious timestep, the anim  
ation revealed the process  
of projecting background p  
oints onto the reference v  
iew, culminating in the im  
age `I12B_img` (the combin  
ed background projectio  
n).\n    - The current time  
step reveals `Mprime1_img`  
(the refined inpainting ma  
sk).\n    - According to th  
e XML, there is a direct e  
dge `arrow_I12B_Mprime1` f  
rom `I12B_img` to `Mprime1  
_img`.\n    - The caption e  
xplicitly states that \"Ex  
cluding these visible back  
ground pixels refines the  
original mask into a small  
er inpainting mask,\" whic  
h perfectly describes the  
transition from the projec  
ted pixels (`I12B_img`) to  
the resulting mask (`Mprim  
e1_img`).\n    - This is th  
e logically correct next s  
tep in the \"Inferring Dep  
th-Guided Inpainting Mask
```

```

\" sequence.\n\n2. **Analyze Criterion B (Group Coherence):**\n    - Only one primary element is revealed in this step: `Mprime1_img` (and its associated label `Mprime1_lbl` which is visually tied to it).\n    - Since it is a single logical unit following a direct dependency from the previous step, there are no grouping conflicts or unrelated elements mixed in. The group is perfectly coherent.\n\n3. **Combining A and B:**\n    - Criterion A is 4 (Fully Sensible).\n    - Criterion B is 4 (Fully Sensible).\n    - The final band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",
    "timestep": 8,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/8_CVPR_2025_pipe00004_frames_tikz_AM.png"
  },
  {
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n

```

- The previous timestep concluded the \"Inferring Depth-Guided Inpainting Mask\" phase, ending with the reveal of `Mprime1\_img`.\n

- The current timestep transitions to the \"Inpainting-guided 3DGS Refinement\" phase.\n

- The newly revealed elements are `Dprime1\_img` (the rendered depth map of the reference view) and `Iprime1\_img` (the rendered image of the reference view), along with the associated label `Dprime1\_lbl\_right` and `Iprime1\_lbl\_right`.\n

- According to the XML, these are the outputs of the refinement process (`arrow\_render\_ref` from `Gprime\_box` to `Iprime1\_img`).\n

- The caption explicitly states: \"the reference view renders its current image and depth map.\"\n

- This is a logical progression: after establishing the mask, the system now shows the current state of the reference view before the inpainting/refinement loop begins.\n

- Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Co

herence):**\n - The elements revealed together are `Dprime1\_img` and `Iprime1\_img` (and their labels).\n - These two elements are structurally and semantically linked: they represent the same view (the reference view ξ_1) but in different modalities (depth vs. RGB).\n - In the XML, they are spatially adjacent and share a combined raster representation `Dprime1\_Iprime1\_img\_r`.\n - There is no prerequisite dependency between the depth map and the image that would require them to be shown in separate steps; they are simultaneous outputs of the rendering process.\n - Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n - Both Criterion A and Criterion B are scored as 4. Following the tie-break rule, the final band is 4.",
"criteria_a_band":
4,
"criteria_b_band":
4,
"final_band": 4,
"label": "Fully Sensible",

```

        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious timestep concluded th
e \"Inferring Depth-Guided
Inpainting Mask\" phase, e
nding with the reveal of `
Mprime1_img`.\n    - The cu
rrent timestep transitions
to the \"Inpainting-guided
3DGS Refinement\" phase.\n
    - The newly revealed eleme
nts are `Dprime1_img` (the
rendered depth map of the
reference view) and `Iprim
e1_img` (the rendered imag
e of the reference view),
along with the associated
label `Dprime1_lbl_right`
and `Iprime1_lbl_right`.\n
    - According to the XML, th
ese are the outputs of the
refinement process (`arrow
_render_ref` from `Gprime
_box` to `Iprime1_img`).\n
    - The caption explicitly s
tates: \"the reference vie
w renders its current imag
e and depth map.\".\n    - T
his is a logical progressi
on: after establishing the
mask, the system now shows
the current state of the r
eference view before the i

```

```

npainting/refinement loop
begins.\n    - Criterion A
is fully satisfied.\n\n2.
**Analyze Criterion B (Gro
up Coherence):**\n    - The
elements revealed together
are `Dprime1_img` and `Ipr
ime1_img` (and their label
s).\n    - These two elemen
ts are structurally and se
mantically linked: they re
present the same view (the
reference view  $\xi_1$ ) b
ut in different modalities
(depth vs. RGB).\n    - In
the XML, they are spatiall
y adjacent and share a com
bined raster representatio
n `Dprime1_Iprime1_img_r
`.\n    - There is no prere
quisite dependency between
the depth map and the imag
e that would require them
to be shown in separate st
eps; they are simultaneous
outputs of the rendering p
rocess.\n    - Criterion B
is fully satisfied.\n\n3.
**Combining A and B:**\n
- Both Criterion A and Cri
terion B are scored as 4.
Following the tie-break ru
le, the final band is 4.\n
\n### CRITERION_A_BAND: 4
\n### CRITERION_B_BAND: 4
\n### FINAL_BAND: 4\n### L

```

```

ABEL: Fully Sensible",
    "timestep": 9,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/9_CVPR_2025_pipe00004_frames_tikz_AM.png"
},
{
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n
- The previous frame ended with the reference view rendering its current image and depth map (`Iprime1_img` and `Dprime1_img`).\n
- The current frame reveals the mask `Mprime1_small_img` and the edge `arrow_mask_flow` (the dashed arrow) connecting the previously inferred mask `Mprime1_img` to this smaller mask.\n
- According to the XML, `Mprime1_img` is the source of `arrow_mask_flow` and `Mprime1_small_img` is the target.\n
- The caption explicitly states: \"The inferred depth-guided mask is forwarded to direct the 2D inpainting process.\".\n
- This is a logically correct progression. The mask was inferred i

```

```

n the \"Inferring Depth-Gu
ided Inpainting Mask\" sec
tion and is now being pass
ed into the \"Inpainting-g
uided 3DGS Refinement\" se
ction to guide the inpaint
ing of the reference vie
w.\n    - Criterion A is fu
lly satisfied.\n\n2. **Ana
lyze Criterion B (Group Co
herence):**\n    - The elem
ents revealed together are
the node `Mprime1_small_im
g` and the edge `arrow_mas
k_flow`.\n    - These two e
lements are intrinsically
linked: one is the target
of the other. They represe
nt a single logical action
(the forwarding of the mas
k).\n    - There are no dep
endency violations; the so
urce node `Mprime1_img` wa
s already revealed in prev
ious steps.\n    - Criterio
n B is fully satisfied.\n
\n3. **Combining A and B:*
*\n    - Both Criterion A a
nd Criterion B are scored
as 4. Following the tie-br
eak rule, the final band i
s 4.",
        "criterion_a_band":
4,
        "criterion_b_band":
4,

```

```

        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous frame ended with the reference view rendering its current image and depth map (`Iprime1_img` and `Dprime1_img`).\n    - The current frame reveals the mask `Mprime1_small_img` and the edge `arrow_mask_flow` (the dashed arrow) connecting the previously inferred mask `Mprime1_img` to this smaller mask.\n    - According to the XML, `Mprime1_img` is the source of `arrow_mask_flow` and `Mprime1_small_img` is the target.\n    - The caption explicitly states: \"The inferred depth-guided mask is forwarded to direct the 2D inpainting process.\"\n    - This is a logically correct progression. The mask was inferred in the \"Inferring Depth-Guided Inpainting Mask\" section and is now being passed into the \"Inpainting-guided 3DGS R

```

```

efinement\" section to guide the inpainting of the reference view.\n
- Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n
- The elements revealed together are the node `Mprime1_small_img` and the edge `arrow_mask_flow`.\n
- These two elements are intrinsically linked: one is the target of the other. They represent a single logical action (the forwarding of the mask).\n
- There are no dependency violations; the source node `Mprime1_img` was already revealed in previous steps.\n
- Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n
- Both Criterion A and Criterion B are scored as 4. Following the tie-break rule, the final band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",
    "timestep": 10,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/10_CVPR_2025_pipe00004_fra

```

```
mes_tikz_AM.png"
    },
    {
        "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- In the previous timeste
p, the mask `Mprime1_small
_img` was revealed, and th
e caption stated it is for
warded to direct the 2D in
painting process.\n
- In the current timestep, the
following elements are rev
ealed:\n
- `DIn1_img`
(the rendered reference im
age and depth map inpu
t).\n
- `arrow_mask_to
_inpaint` (the edge from t
he mask to the inpainting
process).\n
- The text
`"Inpaint\"` (associated wi
th the `arrow_inpaint` or
the process itself).\n
- The caption explicitly men
tions `"A 2D inpainter fil
ls missing regions across
both the rendered referenc
e image and its depth ma
p.`\n
- Revealing `DIn1
_img` now is logically cor
rect as it is the target o
f the mask's influence and
the subject of the current
caption. The reveal follow
s the flow from the mask
```



```

(`Mprime1_small_img`) to the input image (`DIn1_img`).\n\n2. **Analyze Criterion B (Group Coherence):**\n\n    - The elements revealed together are the input image `DIn1_img` and the directional indicator `"Input\` / `arrow_mask_to_inpaint`.\n\n    - These are structurally and logically linked: the mask is being applied to this specific image to perform the inpainting operation.\n\n    - There are no dependency violations; the mask (prerequisite) was already revealed in the previous step. The grouping is coherent as it represents the `"input\` side of the inpainting operation.\n\n3. **Combining A and B:**\n\n    - Criterion A is fully sensible (4): the elements match the caption and the logical flow of the diagram.\n\n    - Criterion B is fully sensible (4): the elements are logically and spatially related.\n\n    - Final Band: 4.",
        "criterion_a_band":
4,
        "criterion_b_band":
4,

```

```

        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n    - In the previous timestep, the mask `Mprime1_small_img` was revealed, and the caption stated it is forwarded to direct the 2D inpainting process.\n    - In the current timestep, the following elements are revealed:\n        - `DIn1_img` (the rendered reference image and depth map input).\n        - `arrow_mask_to_inpaint` (the edge from the mask to the inpainting process).\n        - The text `\"Inpaint\"` (associated with the `arrow_inpaint` or the process itself).\n    - The caption explicitly mentions `\"A 2D inpainter fills missing regions across both the rendered reference image and its depth map.\"`\n    - Revealing `DIn1_img` now is logically correct as it is the target of the mask's influence and the subject of the current caption. The rev

```

eal follows the flow from the mask (`Mprime1\_small\_img`) to the input image (`DIn1\_img`).\n\n2. **Analyze Criterion B (Group Coherence):**\n - The elements revealed together are the input image `DIn1\_img` and the directional indicator `"Inpaint\" / `arrow_mask_to_inpaint`. \n - These are structurally and logically linked: the mask is being applied to this specific image to perform the inpainting operation.\n - There are no dependency violations; the mask (prerequisite) was already revealed in the previous step. The grouping is coherent as it represents the "input" side of the inpainting operation.\n\n3. **Combining A and B:**\n - Criterion A is fully sensible (4): the elements match the caption and the logical flow of the diagram.\n - Criterion B is fully sensible (4): the elements are logically and spatially related.\n - Final Band: 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n#

```

## LABEL: Fully Sensible",
    "timestep": 11,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/11_CVPR_2025_pipe00004_frames_tikz_AM.png"
},
{
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n
- The previous frame focused on the inpainting process (`IIn1_img`, `DIn1_img`, `Mprime1_small_img`).\n
- The current frame reveals the image stack `IP_stack_1` (the \"Project to Other Views\" target) and the edge `arrow_proj_other` connecting the 3D point cloud `P1_box` to this stack.\n
- The caption explicitly states: \"These 3D points are re-projected onto other camera views to form supervisory targets.\"\n
- This is a logical progression. After establishing the inpainted reference view, the animation now shows how the refined 3D geometry (`P1_box`) is used to generate the targets for the cross-view loss

```

```

(`loss_cross`).\n    - The
choice of elements (`IP_st
ack_1` and `arrow_proj_oth
er`) perfectly matches the
semantic intent of the cap
tion and the flow of the d
iagram.\n\n2. **Analyze Cr
iterion B (Group Coherenc
e):**\n    - The elements r
evealed together are the t
arget image stack (`IP_sta
ck_1`) and the projection
arrow (`arrow_proj_other
`).\n    - These have a dir
ect structural and logical
relationship: the arrow re
presents the operation (pr
ojection) and the image st
ack represents the result
of that operation.\n    - T
here are no dependency vio
lations; the source of the
arrow (`P1_box`) was alrea
dy revealed in previous st
eps.\n    - The group is hi
ghly coherent.\n\n3. **Com
bining A and B:**\n    - Cr
iterion A is 4 (Fully Sens
ible).\n    - Criterion B i
s 4 (Fully Sensible).\n
- Final Band is 4.",
    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,

```

```

        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous frame focused on the inpainting process (`IIn1_img`, `DIn1_img`, `Mprime1_small_img`).\n    - The current frame reveals the image stack `IP_stack_1` (the \"Project to Other Views\" target) and the edge `arrow_proj_other` connecting the 3D point cloud `P1_box` to this stack.\n    - The caption explicitly states: \"These 3D points are re-projected onto other camera views to form supervisory targets.\"\n    - This is a logical progression. After establishing the inpainted reference view, the animation now shows how the refined 3D geometry (`P1_box`) is used to generate the targets for the cross-view loss (`loss_cross`).\n    - The choice of elements (`IP_stack_1` and `arrow_proj_other`) perfectly matches the semantic intent of the caption and th

```

```

e flow of the diagram.\n\n
2. **Analyze Criterion B
(Group Coherence):**\n    -
The elements revealed toge
ther are the target image
stack (`IP_stack_1`) and t
he projection arrow (`arro
w_proj_other`).\n    - Thes
e have a direct structural
and logical relationship:
the arrow represents the o
peration (projection) and
the image stack represents
the result of that operati
on.\n    - There are no dep
endency violations; the so
urce of the arrow (`P1_box
`) was already revealed in
previous steps.\n    - The
group is highly coheren
t.\n\n3. **Combining A and
B:**\n    - Criterion A is
4 (Fully Sensible).\n    -
Criterion B is 4 (Fully Se
nsible).\n    - Final Band
is 4.\n\n### CRITERION_A_B
AND: 4\n### CRITERION_B_BA
ND: 4\n### FINAL_BAND: 4\n
### LABEL: Fully Sensibl
e",
        "timestep": 12,
        "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/
13_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"

```

```
},  
{
```

```
    "rationale": "1. **A  
    nalyze Criterion A (Elemen  
    t Appropriateness):**\n    - In the previous timeste  
    p, the animation focused o  
    n the projection of 3D poi  
    nts to other views (`Pl_bo  
    x` $\rightarrow$ `IP_stac  
    k_1`).\n    - In the curren  
    t timestep, the newly reve  
    aled elements are the `arr  
    ow_init` (the blue arrow l  
    abeled \"Initialize\") and  
    the `Gprime_box` (the bloc  
    k containing the refined G  
    aussians).\n    - According  
    to the XML, `arrow_init` c  
    onnects `Pl_box` to `Gprim  
    e_box`. Since `Pl_box` was  
    already revealed (though f  
    aded in the previous fram  
    e, it was the source of th  
    e previous action), initia  
    lizing the new Gaussians f  
    rom that point cloud is th  
    e logical next step in the  
    pipeline.\n    - The reveal  
    is appropriate as it follo  
    ws the flow of the \"Inpai  
    nting-guided 3DGS Refineme  
    nt\" section.\n\n2. **Anal  
    yze Criterion B (Group Coh  
    erence):**\n    - The eleme  
    nts revealed together are
```



```

the edge `arrow_init` and
its target node `Gprime_box`.
- This is a highly coherent group: an edge and the node it points to, representing a single logical transition ("Initialize").
- There are no dependency violations; the source (`P1_box`) was already present.

3. **Combining A and B:**
- Criterion A is fully sensible (Band 4).
- Criterion B is fully coherent (Band 4).
- Final Band is 4.

{
  "criterion_a_band": 4,
  "criterion_b_band": 4,
  "final_band": 4,
  "label": "Fully Sensible",
  "is_valid": true,
  "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n- In the previous timestep, the animation focused on the projection of 3D points to other views (`P1_box` $\rightarrow$ `IP_stack_1`).\n- In the current timestep, the newly revealed element

```

s are the ``arrow_init`` (the blue arrow labeled `\\"Initialize\\"`) and the ``Gprime_box`` (the block containing the refined Gaussians).

- According to the XML, ``arrow_init`` connects ``P1_box`` to ``Gprime_box``. Since ``P1_box`` was already revealed (though faded in the previous frame, it was the source of the previous action), initializing the new Gaussians from that point cloud is the logical next step in the pipeline.
- The reveal is appropriate as it follows the flow of the `\\"Inpainting-guided 3DGS Refinement\\"` section.

2. **Analyze Criterion B (Group Coherence):**

- The elements revealed together are the edge ``arrow_init`` and its target node ``Gprime_box``.
- This is a highly coherent group: an edge and the node it points to, representing a single logical transition (`\\"Initialize\\"`).
- There are no dependency violations; the source (``P1_box``) was already present.

3. **Combining A and B:**

- Criterion A is

```

fully sensible (Band 4).\n
- Criterion B is fully coherent (Band 4).\n
- Final Band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",
    "timestep": 13,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_AM_TikZ/frames/14_CVPR_2025_pipe00004_frames_tikz_AM.png"
},
{
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n
- The previous frame showed the initialization of the updated 3D Gaussians (`Gprime_box`).\n
- The current frame reveals the result of rendering from these updated Gaussians: the stack of novel views (`Iprime_stack_1`, `Iprime_stack_2`, `Iprime_stack_3`), the corresponding label (`Iprime_lbl`), and the edge connecting the Gaussians to these views (`arrow_render_other`).\n
- This logically follows the flow of the diagram: `Gprime_box`

```

$\xrightarrow{\text{arrow_render_other}} \text{Iprime_stack}$.
 - The caption "Novel views of the object-removed scene are then rendered from the updated 3D Gaussians" perfectly matches the revealed elements.
 - Criterion A is fully satisfied.
 2. **Analyze Criterion B (Group Coherence):**
 - The elements revealed together are: the target node (`Iprime_stack`), its label (`Iprime_lbl`), and the connecting edge (`arrow_render_other`).
 - These are structurally and logically tied together as a single "action" (rendering views from the model).
 - There are no prerequisite dependency violations within this group; the edge and the target node are revealed simultaneously, which is standard for a "flow" animation.
 - Criterion B is fully satisfied.
 3. *Combining A and B:*
 - Both Criterion A and Criterion B are scored as 4. According to the tie-break rule, the final band is 4.

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sens
ible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious frame showed the init
ialization of the updated
3D Gaussians (`Gprime_box
`).\n    - The current fram
e reveals the result of re
ndering from these updated
Gaussians: the stack of no
vel views (`Iprime_stack_1
`, `Iprime_stack_2`, `Ipri
me_stack_3`), the correspo
nding label (`Iprime_lbl
`), and the edge connectin
g the Gaussians to these v
iews (`arrow_render_other
`).\n    - This logically f
ollows the flow of the dia
gram: `Gprime_box` $\xrig
htarrow{arrow\\_render\\_o
ther}$ `Iprime_stack`.\n
    - The caption \"Novel view
s of the object-removed sc
ene are then rendered from
the updated 3D Gaussians\"
perfectly matches the reve

```

```

aled elements.\n    - Crite
rion A is fully satisfie
d.\n\n2. **Analyze Criteri
on B (Group Coherence):**
\n    - The elements reveal
ed together are: the targe
t node (`Iprime_stack`), i
ts label (`Iprime_lbl`), a
nd the connecting edge (`a
rrow_render_other`).\n    -
These are structurally and
logically tied together as
a single \"action\" (rende
ring views from the mode
l).\n    - There are no pre
requisite dependency viola
tions within this group; t
he edge and the target nod
e are revealed simultaneou
sly, which is standard for
a \"flow\" animation.\n
- Criterion B is fully sat
isfied.\n\n3. **Combining
A and B:**\n    - Both Crit
erion A and Criterion B ar
e scored as 4. According t
o the tie-break rule, the
final band is 4.\n\n### CR
ITERION_A_BAND: 4\n### CRI
TERION_B_BAND: 4\n### FINA
L_BAND: 4\n### LABEL: Full
y Sensible",
        "timestep": 14,
        "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_AM_TikZ/frames/

```

15_CVPR_2025_pipe00004_frames_tikz_AM.png"

},

{

"rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n

- The previous frame showed the final rendering of novel views and the \"Inpainting-guided 3DGS Refinement\" section.\n - In the current frame, the newly revealed elements are the two loss edges: `loss\_cross` (the red double-headed arrow between $I_{2:K}^P$ and $I'_{2:K}$) and `loss\_render` (the red double-headed arrow between D_1^I and D'_1).\n - The caption explicitly mentions \"rendering losses and cross-view consistency losses optimize the updated 3D Gaussians.\" \n - These elements are the exact visual representations of the losses described in the text. Since this is the final timestep (15/15), revealing the optimization constraints that tie the previous components together is logically appropriate.\n\n2. **Analyze Criterion B

(Group Coherence):**\n - The elements revealed together are `loss\_cross` and `loss\_render`.\n - Both are `"loss\` edges (indicated by the $\mathcal{L}$ notation in the original diagram) and both are styled identically (red double-headed arrows).\n - They are semantically related as the objective functions used for the optimization process described in the caption.\n - There is no prerequisite dependency between these two specific losses; they act as parallel constraints on the optimization of the 3D Gaussians.\n\n3. **Combining A and B:**\n - Criterion A is fully satisfied (4): The elements match the caption and complete the diagram's logic.\n - Criterion B is fully satisfied (4): The elements are semantically and visually coherent.\n - Final Band: 4.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sens

```



```

ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious frame showed the fina
l rendering of novel views
and the \"Inpainting-guide
d 3DGS Refinement\" sectio
n.\n    - In the current fr
ame, the newly revealed el
ements are the two loss ed
ges: `loss_cross` (the red
double-headed arrow betwee
n  $I_{2:K}^P$  and  $I'_{2:K}$ ) and `loss_render` (th
e red double-headed arrow
between  $D_1^{In}$  and
 $D'_1$ ).\n    - The caption
explicitly mentions \"rend
ering losses and cross-vie
w consistency losses optim
ize the updated 3D Gaussia
ns.\"\n    - These elements
are the exact visual repre
sentations of the losses d
escribed in the text. Sinc
e this is the final timest
ep (15/15), revealing the
optimization constraints t
hat tie the previous compo
nents together is logicall
y appropriate.\n\n2. **Ana
lyze Criterion B (Group Co
herence):**\n    - The elem

```

ents revealed together are ``loss_cross`` and ``loss_render``.
 - Both are ``loss`` edges (indicated by the $\mathcal{L}$ notation in the original diagram) and both are styled identically (red double-headed arrows).
 - They are semantically related as the objective functions used for the optimization process described in the caption.
 - There is no prerequisite dependency between these two specific losses; they act as parallel constraints on the optimization of the 3D Gaussians.

3. **Combining A and B:**
 - Criterion A is fully satisfied (4): The elements match the caption and complete the diagram's logic.
 - Criterion B is fully satisfied (4): The elements are semantically and visually coherent.

Final Band: 4.

CRITERION_A_BAND: 4
 CRITERION_B_BAND: 4
 FINAL_BAND: 4
 LABEL: Fully Satisfiable",

"timestep": 15,
 "frame_path": "/archive/medha.sen/DAG_test/ben

```

ch/sample1_AM_TikZ/frames/
16_CVPR_2025_pipe00004_fra
mes_tikz_AM.png"
    }
  ]
}

```

Results for Progressive Reveal

attachment:5177c180-7335-473c-aa5a-8894e11ecd82:CVPR_2025_pipe00010_video_tikz_PR.mp4

```

{
  "sample_dir": "/archive/
medha.sen/DAG_test/bench/s
ample1_PR_TikZ",
  "model": "google/gemma-4
-31B-it",
  "n_timesteps": 16,
  "timesteps": [
    {
      "rationale": "***Crit
erion A: Element Appropria
teness**\n
The animation be
gins by revealing the inpu
t data of the pipeline. Th
e elements revealed are:\n
- `views_lbl` (\n
Views:
 $\xi_{1:K}$ )\n
- `I_1_K_
lbl` and `I_stack_1` (the
first image in the input s
tack)\n
- `M_1_K_lbl` and `
M_stack_1` (the first mask
in the input stack)\n
- The
bottom bracket `bracket1`
and the corresponding capt
ion.\n\n
This is a highly s

```

ensible entry point. The pipeline logically starts with the input images and masks. While the XML defines stacks of images (`I\_stack\_1`, `I\_stack\_2`, `I\_stack\_3`), revealing only the front-most image of each stack is a common and acceptable animation choice to represent the set without cluttering the first frame. No critical prerequisite elements are missing for this introductory step.

Criterion B: Group Coherence

The elements revealed together are the input image stack and the input mask stack. These are semantically and logically linked as they represent the paired input data (image and its corresponding object mask) required for the process. There are no dependency violations here; the labels and their respective images are revealed simultaneously, which is considered a trivial dependency. The grouping is coherent and reflects the "input" phase of the diagram.

Combined Band

Both Criterion A and Criterion

```

ion B are fully satisfied.
The choice of entry point
is logical, and the groupi
ng of images and masks is
coherent.",
    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sens
ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n**Criterion A: E
lement Appropriateness**\n
The animation begins by re
vealing the input data of
the pipeline. The elements
revealed are:\n- `views_lb
l` (\nViews: $\xi_{1:K}$
$\")\n- `I_1_K_lbl` and `I
_stack_1` (the first image
in the input stack)\n- `M
_1_K_lbl` and `M_stack_1`
(the first mask in the inp
ut stack)\n- The bottom br
acket `bracket1` and the c
orresponding caption.\n\nT
his is a highly sensible e
ntry point. The pipeline l
ogically starts with the i
nput images and masks. Whi
le the XML defines stacks
of images (`I_stack_1`, `I
_stack_2`, `I_stack_3`), r

```

revealing only the front-most image of each stack is a common and acceptable animation choice to represent the set without cluttering the first frame. No critical prerequisite elements are missing for this introductory step.

****Criterion B: Group Coherence****

The elements revealed together are the input image stack and the input mask stack. These are semantically and logically linked as they represent the paired input data (image and its corresponding object mask) required for the process. There are no dependency violations here; the labels and their respective images are revealed simultaneously, which is considered a trivial dependency. The grouping is coherent and reflects the "input" phase of the diagram.

****Combined Band****

Both Criterion A and Criterion B are fully satisfied. The choice of entry point is logical, and the grouping of images and masks is coherent.

CRITERION_A_BAND: 4\n##
CRITERION_B_BAND: 4\n###

```

FINAL_BAND: 4\n### LABEL:
Fully Sensible",
    "timestep": 1,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
1_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
        "rationale": "**Crit
erion A: Element Appropria
teness**\nIn the previous
timestep, the input images
(`I_stack_1`) and masks (`
M_stack_1`) were revealed.
The current timestep revea
ls the 3D Gaussian Splatti
ng model block (`G_box` an
d its image `G_box_img`) a
nd the connecting edge (`a
rrow_I_G`) from the input
images to the model. This
is the logically correct n
ext step in the pipeline,
as the caption explicitly
states that the model is
\"pretrained on these mult
i-view input images.\" The
reveal follows the flow of
the diagram from inputs to
the first processing stag
e.\n\n**Criterion B: Group
Coherence**\nThe elements
revealed together are the
target node (`G_box`) and

```

the edge (``arrow_I_G``) that connects the previously revealed source (``I_stack_1``) to this target. This is a highly coherent group as it represents a single logical transition in the data flow. There are no dependency violations; the source of the arrow was already present, and the target is revealed simultaneously with the arrow.\n\n**Combined Band**\nBoth Criterion A and Criterion B are fully satisfied. The reveal is logically sequenced and structurally coherent."

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n**Criterion A: Element Appropriateness**\nIn the previous timestep, the input images (`I_stack_1`) and masks (`M_stack_1`) were revealed. The current timestep reveals the 3D Gaussian Splatting model

```


block (`G\_box` and its image `G\_box\_img`) and the connecting edge (`arrow\_I\_G`) from the input images to the model. This is the logically correct next step in the pipeline, as the caption explicitly states that the model is "pretrained on these multi-view input images." The reveal follows the flow of the diagram from inputs to the first processing stage.

Criterion B: Group Coherence

The elements revealed together are the target node (`G\_box`) and the edge (`arrow\_I\_G`) that connects the previously revealed source (`I\_stack\_1`) to this target. This is a highly coherent group as it represents a single logical transition in the data flow. There are no dependency violations; the source of the arrow was already present, and the target is revealed simultaneously with the arrow.

Combined B and

Both Criterion A and Criterion B are fully satisfied. The reveal is logically sequenced and structurally coherent.

C

```

RITERION_A_BAND: 4\n### CR
ITERION_B_BAND: 4\n### FIN
AL_BAND: 4\n### LABEL: Ful
ly Sensible",
    "timestep": 2,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
2_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
    },
    {
        "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- **Previously revealed:**
`views_lbl`, `I_1_K_lbl`,
`I_stack_1` (and its imag
e), `M_1_K_lbl`, `M_stack_
1` (and its image), `G_box
` (and its image), and the
first bracket `bracket1
`.\n
- **Newly revealed
in current frame:** The ed
ge `arrow_G_D` (`"Render D
epth`"), the text label `D
_1_K_lbl`, and the raster
node `D_stack_1` (and its
image).\n
- **Logic:** T
he flow of the diagram sta
rts with input views and m
asks, which are used to tr
ain the initial Gaussian S
platting model (`G_box`).
The next logical step in t
he pipeline is to use that

```

model to render depth maps. The reveal of the `\Render Depth\` arrow and the resulting depth stack ``D_stack_1`` follows the structural flow defined by the edge ``arrow_G_D`` perfectly.

- ****Verdict:**** Criterion A is fully satisfied.

2. ****Analyze Criterion B (Group Coherence):****

- ****Elements grouped:**** ``arrow_G_D``, ``D_1_K_lbl``, and ``D_stack_1``.

- ****Relation:**** These elements are intrinsically linked. The arrow represents the process, the label identifies the output, and the raster node is the visual representation of that output. They form a single logical unit of information (the `\Render Depth\` operation).

- ****Dependency:**** There is a prerequisite dependency where ``G_box`` must exist before ``D_stack_1`` can be rendered from it; since ``G_box`` was revealed in the previous step, this dependency is respected.

- ****Verdict:**** Criterion B is fully satisfied.

3. ****Combining Bands:****

- Criterion A: 4

```

\n    - Criterion B: 4\n
    - Both criteria are fully
    sensible.",
        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sens
ible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - **Previo
usly revealed:** `views_lb
l`, `I_1_K_lbl`, `I_stack_
1` (and its image), `M_1_K
_lbl`, `M_stack_1` (and it
s image), `G_box` (and its
image), and the first brac
ket `bracket1`.\n    - **Ne
wly revealed in current fr
ame:** The edge `arrow_G_D
` (`"Render Depth`"), the
text label `D_1_K_lbl`, an
d the raster node `D_stack
_1` (and its image).\n    -
**Logic:** The flow of the
diagram starts with input
views and masks, which are
used to train the initial
Gaussian Splatting model
(`G_box`). The next logica
l step in the pipeline is
to use that model to rende

```

r depth maps. The reveal of the \"Render Depth\" arrow and the resulting depth stack `D\_stack\_1` follows the structural flow defined by the edge `arrow\_G\_D` perfectly.

- **Verdict:** Criterion A is fully satisfied.

2. **Analyze Criterion B (Group Coherence):**

- **Elements grouped:** `arrow\_G\_D`, `D\_1\_K\_lbl`, and `D\_stack\_1`.

- **Relation:** These elements are intrinsically linked. The arrow represents the process, the label identifies the output, and the raster node is the visual representation of that output. They form a single logical unit of information (the \"Render Depth\" operation).

- **Dependency:** There is a prerequisite dependency where `G\_box` must exist before `D\_stack\_1` can be rendered from it; since `G\_box` was revealed in the previous step, this dependency is respected.

- **Verdict:** Criterion B is fully satisfied.

3. **Combining Bands:**

- **Criterion A:** 4

- **Criterion B:**

```

4\n    - Both criteria are
fully sensible.\n\n### CRI
TERION_A_BAND: 4\n### CRIT
ERION_B_BAND: 4\n### FINAL
_BAND: 4\n### LABEL: Fully
Sensible",
    "timestep": 3,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
3_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
},
{
    "rationale": "**Crit
erion A: Element Appropria
teness**\n\nThe previous tim
estep established the init
ial state: the input views
( $I_{1:K}$ ), masks ( $M_{1:K}$ ), the Gaussian represe
ntation ( $G_{box}$ ), and t
he rendered depth maps ( $D_{1:K}$ ). The current time
step reveals the transitio
n to the next logical phas
e: \"Inferring Depth-Guide
d Inpainting Mask\". Speci
fically, it reveals the in
dividual reference views a
nd masks: `box_I2_M2` (con
taining `I2_lbl`, `I2_img`
`, `M2_lbl`, `M2_img`) and
`box_I1_M1` (containing `I
1_lbl`, `I1_img`, `M1_lbl`
`, `M1_img`), along with t

```

he corresponding bracket label `bracket2`. This is a logically sound progression, moving from the general stacks of images to the specific pairs used for the inpainting mask inference.

Criterion B: Group Coherence

The elements revealed together are highly coherent. `box\_I2\_M2` and `box\_I1\_M1` are parallel structural units in the XML, both representing the same semantic concept (a pair of an image and its mask for a specific view). They are spatially adjacent and logically grouped under the "Inferring Depth-Guided Inpainting Mask" phase. There are no prerequisite dependency violations; these are the starting inputs for the subsequent operations (like the arrows leading to I^B and D_2 which are not yet revealed).

Combined Band

Both Criterion A and Criterion B are fully satisfied. The reveal follows the diagram's flow and groups related structural elements together.",

"criterion_a_band":

```

4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sens
ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n**Criterion A: E
lement Appropriateness**\n
The previous timestep esta
blished the initial state:
the input views ($I_{1:K}
$), masks ($M_{1:K}$), the
Gaussian representation
($G_{box}$), and the rende
red depth maps ($D_{1:K}
$). The current timestep r
eveals the transition to t
he next logical phase: \
Inferring Depth-Guided Inpa
inting Mask\". Specificall
y, it reveals the individu
al reference views and mas
ks: `box_I2_M2` (containin
g `I2_lbl`, `I2_img`, `M2_
lbl`, `M2_img`) and `box_I
1_M1` (containing `I1_lbl
`, `I1_img`, `M1_lbl`, `M1
_img`), along with the cor
responding bracket label `
bracket2`. This is a logic
ally sound progression, mo
ving from the general stac
ks of images to the specif
ic pairs used for the inpa

```


inting mask inference.\n\n
 Criterion B: Group Coherence\n
 The elements revealed together are highly coherent. `box\_I2\_M2` and `box\_I1\_M1` are parallel structural units in the XML, both representing the same semantic concept (a pair of an image and its mask for a specific view). They are spatially adjacent and logically grouped under the "Inferring Depth-Guided Inpainting Mask" phase. There are no prerequisite dependency violations; these are the starting inputs for the subsequent operations (like the arrows leading to I^B and D_2 which are not yet revealed).\n\n
 Combined Band\n
 Both Criterion A and Criterion B are fully satisfied. The reveal follows the diagram's flow and groups related structural elements together.\n\n
 ### CRITERION_A_BAND: 4\n
 ### CRITERION_B_BAND: 4\n
 ### FINAL_BAND: 4\n
 ### LABEL: Fully Sensible",
 "timestep": 4,
 "frame_path": "/archive/medha.sen/DAG_test/ben

```
ch/sample1_PR_TikZ/frames/  
4_CVPR_2025_pipe00004_fram  
es_tikz_PR_animated.png"
```

```
},
```

```
{
```

```
    "rationale": "1. **A  
analyze Criterion A (Elemen  
t Appropriateness):**\n
```

```
- The previous timestep re  
vealed the input views ( $I_{1:K}$ ), masks ( $M_{1:K}$   
) , the Gaussian model ( $G_{1:N}$ ), and the rendered  
depth ( $D_{1:K}$ ), as well  
as the specific reference  
views  $I_1, I_2$  and masks  
 $M_1, M_2$ . \n
```

```
- In the c  
urrent timestep, the newly  
revealed elements are the  
background-extracted image  
s  $I_1^B$  (I1B_img` and I1B_lbl`) and  $I_2^B$  (I2  
B_img` and I2B_lbl`). \n
```

```
- According to the XML edg  
es, arrow_M1_I1B` connect  
s  $M_1$  to  $I_1^B$ , and arrow_M2_I2B` connects  $M_2$   
 to  $I_2^B$ . Since  $M_1$   
 and  $M_2$  were already rev  
ealed, revealing the resul  
t of applying these masks  
to the images is the logic  
ally correct next step in  
the flow. \n
```

```
- The captio  
n explicitly states: \n "The  
original visible backgroun
```

d regions for each viewpoint are first extracted by applying the negative of the object masks, which perfectly matches the reveal of I_1^B and I_2^B .

- Criterion A is fully satisfied.

2. **Analyze Criterion B (Group Coherence):**

- The elements revealed together are $\{I_1^B, I_1^B \setminus I_1^B\}$ and $\{I_2^B, I_2^B \setminus I_2^B\}$.

- These are parallel operations: the same process (masking) is applied to two different reference views. They are structurally and semantically related as they represent the same stage of the pipeline for different inputs.

- There is no prerequisite dependency between I_1^B and I_2^B ; they can be computed independently.

- Criterion B is fully satisfied.

3. **Combining A and B:**

- Both Criterion A and Criterion B are scored as 4. The combined band is 4.

"criterion_a_band": 4,

"criterion_b_band": 4,

```

        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous timestep revealed the input views ($I_{1:K}$), masks ($M_{1:K}$), the Gaussian model ($G_{1:N}$), and the rendered depth ($D_{1:K}$), as well as the specific reference views $I_1, I_2$ and masks $M_1, M_2$.\n    - In the current timestep, the newly revealed elements are the background-extracted images $I_1^B$ (`I1B_img` and `I1B_lbl`) and $I_2^B$ (`I2B_img` and `I2B_lbl`).\n    - According to the XML edges, `arrow_M1_I1B` connects $M_1$ to $I_1^B$, and `arrow_M2_I2B` connects $M_2$ to $I_2^B$. Since $M_1$ and $M_2$ were already revealed, revealing the result of applying these masks to the images is the logically correct next step in the flow.\n    - The caption explicitly states: \"The original visible background re

```

gions for each viewpoint are first extracted by applying the negative of the object masks, \" which perfectly matches the reveal of \$I_1^B\$ and \$I_2^B\$.\n - Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n - The elements revealed together are \$\{I_1^B, I_1B_l_l\}\$ and \$\{I_2^B, I_2B_l_l\}\$.\n - These are parallel operations: the same process (masking) is applied to two different reference views. They are structurally and semantically related as they represent the same stage of the pipeline for different inputs.\n - There is no prerequisite dependency between \$I_1^B\$ and \$I_2^B\$; they can be computed independently.\n - Criterion B is fully satisfied.\n\n3. **Combining A and B:**\n - Both Criterion A and Criterion B are scored as 4. The combined band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",

```

        "timestep": 5,
        "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/5_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"
    },
    {
        "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n
        - The previous timestep revealed the initial views, masks, and the extraction of background regions ($I_1^B$, $I_2^B$).\n
        - The current timestep reveals:\n
            - `D2_lbl` and `D2_img` (the depth map for the second view).\n
            - `I12B_lbl` and `I12B_img` (the projected background image).\n
            - The edges `arrow_I2B_D2` and `arrow_D2_I12B` (and `arrow_I1B_I12B`).\n
            - This logically follows the flow of the \"Inferring Depth-Guided Inpainting Mask\" section. After extracting background regions, the next step is to use depth maps to project these regions into a common reference view. The elements revealed are exactly those required to illu

```

strate this projection process.\n - Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n - The elements revealed together are `D2\_img` and `I12B\_img`, along with their labels and the connecting arrows.\n - These are structurally and logically linked: `D2\_img` is a prerequisite for the projection that results in `I12B\_img` (as indicated by `edge id= \"arrow\_D2\_I12B\").\n - While `D2_img` is a prerequisite for `I12B_img`, they are part of the same functional operation (the projection step). In the context of a diagram reveal, grouping a node and its immediate successor in a small sub-operation is generally acceptable and coherent, as they together represent the \"projection\" action described in the caption.\n - There are no unrelated elements from other branches mixed in.\n - Criterion B is fully satisfied.\n\n3. **Combining Bands:**\n - Criterion A: 4\n - Criterion B: 4\n

```

- Both criteria are fully
sensible.",
    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sens
ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious timestep revealed the
initial views, masks, and
the extraction of backgrou
nd regions ($I_1^B$, $I_2^
B$).\n    - The current tim
estep reveals:\n        - `D2
_lbl` and `D2_img` (the de
pth map for the second vie
w).\n        - `I12B_lbl` and
`I12B_img` (the projected
background image).\n        -
The edges `arrow_I2B_D2` a
nd `arrow_D2_I12B` (and `a
rrow_I1B_I12B`).\n    - Thi
s logically follows the fl
ow of the \"Inferring Dept
h-Guided Inpainting Mask\"
section. After extracting
background regions, the ne
xt step is to use depth ma
ps to project these region
s into a common reference

```


view. The elements revealed are exactly those required to illustrate this projection process.

- Criterion A is fully satisfied.

2. **Analyze Criterion B (Group Coherence):**

- The elements revealed together are `D2\_img` and `I12B\_img`, along with their labels and the connecting arrows.
- These are structurally and logically linked: `D2\_img` is a prerequisite for the projection that results in `I12B\_img` (as indicated by `edge id="arrow\_D2\_I12B"`).
- While `D2\_img` is a prerequisite for `I12B\_img`, they are part of the same functional operation (the projection step). In the context of a diagram reveal, grouping a node and its immediate successor in a small sub-operation is generally acceptable and coherent, as they together represent the "projection" action described in the caption.
- There are no unrelated elements from other branches mixed in.
- Criterion B is fully satisfied.

3. **Com**

```

binning Bands:**\n    - Crit
erion A: 4\n    - Criterion
B: 4\n    - Both criteria a
re fully sensible.\n\n###
CRITERION_A_BAND: 4\n### C
RITERION_B_BAND: 4\n### FI
NAL_BAND: 4\n### LABEL: Fu
lly Sensible",
    "timestep": 6,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
6_CVPR_2025_pipe00004_fram
es_tikz_PR_animated.png"
},
{
    "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- In the previous timeste
p, the animation had revea
led the flow from the view
s/masks down to the projec
ted background image `I12B
_img`.\n    - The current t
imestep reveals the final
output of the \"Inferring
Depth-Guided Inpainting Ma
sk\" section: the refined
mask `Mprime1_img` and its
associated label `Mprime1_
lbl`.\n    - According to t
he XML, there is a direct
edge `arrow_I12B_Mprime1`
from `I12B_img` to `Mprime
1_img`. Since `I12B_img` w

```

as the last element revealed, `Mprime1\_img` is the logically correct next element to reveal.\n - The reveal is appropriate and follows the structural flow of the diagram.\n\n2. **Analyze Criterion B (Group Coherence):**\n - The elements revealed together are the raster node `Mprime1\_img` and its text label `Mprime1\_lbl`.\n - Per the rubric, grouping a node and its own text label is considered a trivial/instantaneous dependency and does not constitute a violation.\n - These two elements are semantically and structurally linked as a single logical unit.\n\n3. **Combining A and B:**\n - Criterion A is fully sensible (Band 4).\n - Criterion B is fully coherent (Band 4).\n - The final band is 4.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,

```

```
"raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - In the p
revious timestep, the anim
ation had revealed the flo
w from the views/masks dow
n to the projected backgro
und image `I12B_img`.\n
    - The current timestep rev
eals the final output of t
he \"Inferring Depth-Guide
d Inpainting Mask\" sectio
n: the refined mask `Mprim
e1_img` and its associated
label `Mprime1_lbl`.\n    -
According to the XML, ther
e is a direct edge `arrow_
I12B_Mprime1` from `I12B_i
mg` to `Mprime1_img`. Sinc
e `I12B_img` was the last
element revealed, `Mprime1
_img` is the logically cor
rect next element to revea
l.\n    - The reveal is app
ropriate and follows the s
tructural flow of the diag
ram.\n\n2. **Analyze Crite
rion B (Group Coherence):**\n
    - The elements revea
led together are the raste
r node `Mprime1_img` and i
ts text label `Mprime1_lbl
`.\n    - Per the rubric, g
rouping a node and its own
text label is considered a
```

```

trivial/instantaneous dependency and does not constitute a violation.\n
- These two elements are semantically and structurally linked as a single logical unit.\n\n
3. **Combining A and B:**\n
- Criterion A is fully sensible (Band 4).\n
- Criterion B is fully coherent (Band 4).\n
- The final band is 4.\n\n
### CRITERION_A_BAND: 4\n
## CRITERION_B_BAND: 4\n
# FINAL_BAND: 4\n
### LABEL: Fully Sensible",
    "timestep": 7,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/7_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"
},
{
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n
- In the previous timestep, the animation had just revealed the result of the \"Inferring Depth-Guided Inpainting Mask\" section, specifically the mask `Mprime1_img` (and its label `Mprime1_lbl`).\n
- In the current timestep, the an

```

imation reveals `Mprime1\_small\_img` (the smaller version of the mask used in the refinement stage), its label `Mprime1\_small\_lbl`, and the edge `arrow\_mask\_flow` connecting the two.
 - This is logically correct. The diagram flow moves from the mask generation (`Mprime1\_img`) to the refinement stage. The `arrow\_mask\_flow` explicitly links these two elements in the XML. The reveal follows the directional flow of the diagram.
 - Additionally, the bracket label `bracket3` (`"Inpainting-guided 3DGS Refinement"`) is revealed, which correctly introduces the new section of the diagram being entered.
 2. **Analyze Criterion B (Group Coherence):**
 - The elements revealed together are: `Mprime1\_small\_img`, `Mprime1\_small\_lbl`, `arrow\_mask\_flow`, and `bracket3`.
 - These are highly coherent. The image and its label are a single semantic unit. The arrow represents the direct dependency/flow from the previously revealed elemen

t to the new one. The bracket label provides the structural context for the entire right-hand side of the diagram.\n - There are no dependency violations; the source of the arrow (`Mprime1\_img`) was already present.\n\n3. **Combining A and B:**\n - Criterion A is fully sensible (4).\n - Criterion B is fully coherent (4).\n - Final band is 4.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n - In the previous timestep, the animation had just revealed the result of the \"Inferring Depth-Guided Inpainting Mask\" section, specifically the mask `Mprime1_img` (and its label `Mprime1_lb`).\n - In the current timestep, the animation reveals `Mprime1_small_img`

```

(the smaller version of the mask used in the refinement stage), its label ``Mprime1_small_lbl``, and the edge ``arrow_mask_flow`` connecting the two.
- This is logically correct. The diagram flow moves from the mask generation (``Mprime1_img``) to the refinement stage. The ``arrow_mask_flow`` explicitly links these two elements in the XML. The reveal follows the directional flow of the diagram.
- Additionally, the bracket label ``bracket3`` (``("Inpainting-guided 3DGS Refinement")``) is revealed, which correctly introduces the new section of the diagram being entered.
2.
Analyze Criterion B (Group Coherence):
- The elements revealed together are: ``Mprime1_small_img``, ``Mprime1_small_lbl``, ``arrow_mask_flow``, and ``bracket3``.
- These are highly coherent. The image and its label are a single semantic unit. The arrow represents the direct dependency/flow from the previously revealed element to the new one. The bracket label p

rovides the structural context for the entire right-hand side of the diagram.\n - There are no dependency violations; the source of the arrow (`Mprime1\_img`) was already present.\n\n3. **Combining A and B:**\n - Criterion A is fully sensible (4).\n - Criterion B is fully coherent (4).\n - Final band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",

"timestep": 8,
"frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/8_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"

},

{

"rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n - The previous timestep ended with the reveal of `Mprime1\_img` (the inferred inpainting mask) and the caption explaining its purpose.\n - The current timestep reveals `Dprime1\_img` (the rendered depth map fo

r the reference view) and its corresponding label `Dprime1\_lbl\_right`. \n - Looking at the XML and the original diagram, the \"In painting-guided 3DGS Refinement\" stage begins by obtaining the current rendered image (`Iprime1\_img`) and depth map (`Dprime1\_img`) from the scene model. \n - However, the animation reveals `Dprime1\_img` but *omits* `Iprime1\_img` (the rendered image), despite the caption explicitly stating: \"the reference view's current rendered image and depth map are obtained\". \n - Because the caption explicitly mentions both the image and the depth map, but only the depth map is revealed, there is a significant missing element (`Iprime1\_img`) that logically belongs in this step. This is a violation of Criterion A. \n \n 2. **Analyze Criterion B (Group Coherence):** \n - The elements revealed are `Dprime1\_img` and `Dprime1\_lbl\_right`. \n - These are structurally and semantically linked (a node and its label

l), which is a trivial and correct grouping. There are no dependency violations within this small group.\n\n3. **Combining A and B:**\n\n- Criterion A is a Major Violation (Band 1) because a necessary element mentioned in the accompanying text (`Iprime1\_img`) is missing, making the reveal incomplete and contradictory to the caption.\n\n- Criterion B is Fully Sensible (Band 4).\n\n- According to the tie-break rule, I must report the lower band.",

```

        "criterion_a_band":
1,
        "criterion_b_band":
4,
        "final_band": 1,
        "label": "Major Violation",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n\n- The previous timestep ended with the reveal of `Mprime1_img` (the inferred inpainting mask) and the caption explaining its purpose.\n\n- The current timestep reveal

```

s `Dprime1\_img` (the rendered depth map for the reference view) and its corresponding label `Dprime1\_lbl\_right`.\n - Looking at the XML and the original diagram, the \"Inpainting-guided 3DGS Refinement\" stage begins by obtaining the current rendered image (`Iprime1\_img`) and depth map (`Dprime1\_img`) from the scene model.\n - However, the animation reveals `Dprime1\_img` but *omits* `Iprime1\_img` (the rendered image), despite the caption explicitly stating: \"the reference view's current rendered image and depth map are obtained\".\n - Because the caption explicitly mentions both the image and the depth map, but only the depth map is revealed, there is a significant missing element (`Iprime1\_img`) that logically belongs in this step. This is a violation of Criterion A.\n\n2. **Analyze Criterion B (Group Coherence):**\n\n- The elements revealed are `Dprime1\_img` and `Dprime1\_lbl\_right`.\n - These are structurally and

```

semantically linked (a node and its label), which is a trivial and correct grouping. There are no dependency violations within this small group.\n\n3. **Combining A and B:**\n    - Criterion A is a Major Violation (Band 1) because a necessary element mentioned in the accompanying text (`Ip_rime1_img`) is missing, making the reveal incomplete and contradictory to the caption.\n    - Criterion B is Fully Sensible (Band 4).\n    - According to the tie-break rule, I must report the lower band.\n\n### CRITERION_A_BAND: 1\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 1\n### LABEL: Major Violation",
        "timestep": 9,
        "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/9_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"
    },
    {
        "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous frame established the \"refinement st

```

age\" by showing the rendered reference view's current image and depth map (\$I'_1\$ and \$D'_1\$).\n - The current frame reveals the \"inpaint\" operation. Specifically, it adds the input image/depth pair (\$I^{\{In\}}_1\$ and \$D^{\{In\}}_1\$) and the \"Inpaint\" arrow connecting them to the previously revealed \$I'_1\$ and \$D'_1\$.\n - Mapping to XML: The new elements are `IIn1\_img`, `DIn1\_img`, `IIn1\_lbl`, `DIn1\_lbl`, and the `arrow\_inpaint` (which logically connects the reference view to the inpainting process).\n - This is the correct logical progression: after obtaining the reference view, the next step is to apply the 2D inpainting model as described in the caption.\n\n2.

****Analyze Criterion B (Group Coherence):****\n - The elements revealed together are the input pair for the inpainting process (`IIn1\_img`, `DIn1\_img`) and the transition arrow (`arrow\_inpaint`).\n - These are structurally and semantically linked as they represe

```

nt the \"before\" state of
the inpainting operatio
n.\n    - There are no depe
ndency violations; the tar
get of the inpainting
($I'_1, D'_1$) was already
present, and the source el
ements are revealed simult
aneously.\n\n3. **Combinin
g A and B:**\n    - Criteri
on A is fully sensible
(4).\n    - Criterion B is
fully coherent (4).\n    -
Final band is 4.",
        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sens
ible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious frame established the
\"refinement stage\" by sh
owing the rendered referen
ce view's current image an
d depth map ($I'_1$ and
$D'_1$).\n    - The current
frame reveals the \"inpain
t\" operation. Specificall
y, it adds the input imag
e/depth pair ($I^{In}_1$ a

```

nd \$D^{In}_1\$) and the \("I npaint\" arrow connecting them to the previously revealed \$I'_1\$ and \$D'_1\$.\n

- Mapping to XML: The new elements are `IIn1\_img`, `DIn1\_img`, `IIn1\_lbl`, `DIn1\_lbl`, and the `arrow\_inpaint` (which logically connects the reference view to the inpainting process).\n

- This is the correct logical progression: after obtaining the reference view, the next step is to apply the 2D inpainting model as described in the caption.\n\n2. **Analyze Criterion B (Group Coherence):**\n

- The elements revealed together are the input pair for the inpainting process (`IIn1\_img`, `DIn1\_img`) and the transition arrow (`arrow\_inpaint`).\n

- These are structurally and semantically linked as they represent the \("before\" state of the inpainting operation.\n

- There are no dependency violations; the target of the inpainting (\$I'_1\$, \$D'_1\$) was already present, and the source elements are revealed simultaneously.\n


```

\n3. **Combining A and B:*
*\n    - Criterion A is fully
sensible (4).\n    - Cri
terion B is fully coherent
(4).\n    - Final band is
4.\n\n### CRITERION_A_BAN
D: 4\n### CRITERION_B_BAN
D: 4\n### FINAL_BAND: 4\n#
## LABEL: Fully Sensible",
    "timestep": 10,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
10_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    },
    {
        "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- The previous frame showe
d the inpainting process
(the \"Inpaint\" arrow and
the resulting images  $I^{\{I
n\}}_1$  and  $D^{\{In\}}_1$  and t
heir primed counterpart
s).\n    - The current fram
e reveals the loss_render
` edge (the red double-hea
ded arrow labeled  $\mathcal{L}_{render}$ ) connectin
g  $D^{\{In\}}_1$  (DIn1_img`)
and  $D'_1$  (Dprime1_img
`).\n    - Logically, after
the inpainting targets are
generated, the next step i

```

n a training/refinement loop is to define the loss function used to align the rendered outputs with these targets. This is a correct and logical progression in the diagram's flow.

- The caption also updates to describe the rendering loss, which perfectly matches the visual reveal of ``loss_render``.

2. **Analyze Criterion B (Group Coherence):**

- The only new visual element is the ``loss_render`` edge.
- Since only one element is revealed, there are no grouping conflicts. The element is structurally and semantically linked to the nodes it connects, both of which were already present.

3. **Combining A and B:**

- Criterion A is fully sensible (4).
- Criterion B is fully sensible (4).
- The combined band is 4.

```

    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sensible",

```

```

        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious frame showed the inpa
inting process (the \"Inpa
int\" arrow and the result
ing images  $I^{In}_1$  and
 $D^{In}_1$  and their prime
d counterparts).\n    - The
current frame reveals the
`loss_render` edge (the re
d double-headed arrow labe
led  $\mathcal{L}_{render}$ 
) connecting  $D^{In}_1$ 
(`DIn1_img`) and  $D'_1$  (`
Dprime1_img`).\n    - Logic
ally, after the inpainting
targets are generated, the
next step in a training/re
finement loop is to define
the loss function used to
align the rendered outputs
with these targets. This i
s a correct and logical pr
ogression in the diagram's
flow.\n    - The caption al
so updates to describe the
rendering loss, which perf
ectly matches the visual r
eveal of `loss_render`.\n
\n2. **Analyze Criterion B
(Group Coherence):**\n    -
The only new visual elemen
t is the `loss_render` edg

```

```

e.\n    - Since only one element is revealed, there are no grouping conflicts. The element is structurally and semantically linked to the nodes it connects, both of which were already present.\n\n3. **Combining A and B:**\n    - Criterion A is fully sensible (4).\n    - Criterion B is fully sensible (4).\n    - The combined band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n### LABEL: Fully Sensible",
    "timestep": 11,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/11_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"
  },
  {
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous frame had already revealed the inpainting process (`IIn1_img`, `DIn1_img`, `Iprime1_img`, `Dprime1_img`) and the mask flow (`Mprime1_img` $\rightarrow$ `Mprime1_small_img`).\n    - The current f

```

frame reveals the point cloud/anchor point representation `P1\_box` (including its image `P1\_box\_img` and 3D components) and the edge `arrow\_depth\_proj` which connects the depth map `DIn1\_img` to `P1\_box`.\n
 - According to the XML, `arrow\_depth\_proj` is the logical next step in the \"Inpainting-guided 3DGS Refinement\" flow: once the inpainted depth/image is obtained, it is projected into 3D space.\n
 - The reveal is timely and logically follows the flow of the diagram.\n\n
 2. **Analyze Criterion B (Group Coherence):**\n
 - The elements revealed together are the target node `P1\_box` and the incoming edge `arrow\_depth\_proj`.\n
 - This is a highly coherent group as it represents a single logical operation: \"Depth-Guided Projection\".\n
 - There are no dependency violations; the source of the edge (`DIn1\_img`) was already present in the previous frame.\n\n
 3. **Combining A and B:**\n
 - Criterion A is fully sensible (4).\n

```

- Criterion B is fully coherent (4).\n    - Final band is 4.",
    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sensible",
    "is_valid": true,
    "raw_output": "### RATIONALE\n1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous frame had already revealed the inpainting process (`IIn1_img`, `DIn1_img`, `Iprime1_img`, `Dprime1_img`) and the mask flow (`Mprime1_img`  $\rightarrow$  `Mprime1_small_img`).\n    - The current frame reveals the point cloud/anchor point representation `P1_box` (including its image `P1_box_img` and 3D components) and the edge `arrow_depth_proj` which connects the depth map `DIn1_img` to `P1_box`.\n    - According to the XML, `arrow_depth_proj` is the logical next step in the "Inpainting-guided 3DGS Refinement" flow: once the inpai

```

```

nted depth/image is obtained, it is projected into 3
D space.\n    - The reveal
is timely and logically fo
llows the flow of the diag
ram.\n\n2. **Analyze Crite
rion B (Group Coherence):*
*\n    - The elements revea
led together are the targe
t node `P1_box` and the in
coming edge `arrow_depth_p
roj`.\n    - This is a high
ly coherent group as it re
presents a single logical
operation: \"Depth-Guided
Projection\".\n    - There
are no dependency violatio
ns; the source of the edge
(`DIn1_img`) was already p
resent in the previous fra
me.\n\n3. **Combining A an
d B:**\n    - Criterion A i
s fully sensible (4).\n
    - Criterion B is fully coh
erent (4).\n    - Final ban
d is 4.\n\n### CRITERION_A
_BAND: 4\n### CRITERION_B
_BAND: 4\n### FINAL_BAND: 4
\n### LABEL: Fully Sensibl
e",

        "timestep": 12,
        "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
12_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"

```

```

    },
    {
        "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- The previous frame showe
d the projection of the in
painted region into 3D spa
ce (`P1_box`).\n
- The c
urrent frame reveals the p
rojection of these anchor
points back to other train
ing camera views.\n
- Sp
ecifically, the new elemen
ts are: the text label `IP
_lbl` (`"$I_{2:K}^P$`), t
he image stack `IP_stack_1
` (and its associated stac
k layers `IP_stack_2`, `IP
_stack_3`), and the edge `
arrow_proj_other` connecti
ng `P1_box` to `IP_stack_1
`.\n
- This logically fo
llows the flow of the diag
ram: Depth-Guided Projecti
on $\rightarrow$ Anchor P
oints (`P1_box`) $\rightarrow$ Project to Other Vie
ws (`IP_stack_1`). The rev
eal is perfectly timed wit
h the caption provided in
the frame.\n\n2. **Analyze
Criterion B (Group Coheren
ce):**\n
- The elements
revealed together are the
target node (`IP_stack_1

```



```

`), its label (`IP_lbl`),
and the connecting edge (`
arrow_proj_other`).\n    -
This is a highly coherent
group as it represents a s
ingle logical step in the
process (the act of projec
ting and the resulting ima
ges). There are no depende
ncy violations; the source
node `P1_box` was already
revealed in the previous t
imestep.\n\n3. **Combining
A and B:**\n    - Criterion
A is 4 (Fully Sensible).\n
- Criterion B is 4 (Fully
Sensible).\n    - Final ban
d is 4.",
    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sens
ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious frame showed the proj
ection of the inpainted re
gion into 3D space (`P1_bo
x`).\n    - The current fra
me reveals the projection
of these anchor points bac

```

k to other training camera views.
 - Specifically, the new elements are: the text label `IP\_lbl` ($\text{"I}_{2:K}^{\text{P}}$), the image stack `IP\_stack\_1` (and its associated stack layers `IP\_stack\_2`, `IP\_stack\_3`), and the edge `arrow\_proj\_other` connecting `P1\_box` to `IP\_stack\_1`.
 - This logically follows the flow of the diagram: Depth-Guided Projection $\rightarrow$ Anchor Points (`P1\_box`) $\rightarrow$ Project to Other Views (`IP\_stack\_1`). The reveal is perfectly timed with the caption provided in the frame.
 2. **Analyze Criterion B (Group Coherence):**
 - The elements revealed together are the target node (`IP\_stack\_1`), its label (`IP\_lbl`), and the connecting edge (`arrow\_proj\_other`).
 - This is a highly coherent group as it represents a single logical step in the process (the act of projecting and the resulting images). There are no dependency violations; the source node `P1\_box` was already revealed in

```

the previous timestep.\n\n
3. **Combining A and B:**
\n    - Criterion A is 4 (Fully Sensible).\n    - Criterion B is 4 (Fully Sensible).\n    - Final band is 4.\n\n### CRITERION_A_BAND: 4\n### CRITERION_B_BAND: 4\n### FINAL_BAND: 4\n## LABEL: Fully Sensible",
    "timestep": 13,
    "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/13_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"
},
{
    "rationale": "1. **Analyze Criterion A (Element Appropriateness):**\n    - The previous frame showed the point cloud/anchor points in `P1_box` and the resulting inpainting results.\n    - The current frame reveals the `arrow_init` (the blue arrow labeled `\"Initialize\"`) and the `Gprime_box` (the refined 3D Gaussian representation).\n    - According to the XML, `arrow_init` connects `P1_box` to `Gprime_box`. Since `P1_box` was already revealed, it is logically

```

the next step to show how these anchor points are used to initialize the new Gaussians (`Gprime\_box`).

- The reveal is appropriate as it follows the flow of the "Inpainting-guided 3DGS Refinement" section of the diagram.

2. **Analyze Criterion B (Group Coherence):**

- The elements revealed together are the edge `arrow\_init` and the target node `Gprime\_box`.
- This is a highly coherent group: an edge and its destination node. They are structurally and logically linked. There are no dependency violations, as the source node (`P1\_box`) was already present.

3. **Combining A and B:**

- Criterion A is 4 (Fully Sensible).
- Criterion B is 4 (Fully Sensible).
- The combined band is 4.

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sensible",
        "is_valid": true,

```

```
"raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n    - The prev
ious frame showed the poin
t cloud/anchor points in `
P1_box` and the resulting
inpainting results.\n    -
The current frame reveals
the `arrow_init` (the blue
arrow labeled \"Initialize
\") and the `Gprime_box`
(the refined 3D Gaussian r
epresentation).\n    - Acco
rding to the XML, `arrow_i
nit` connects `P1_box` to
`Gprime_box`. Since `P1_bo
x` was already revealed, i
t is logically the next st
ep to show how these ancho
r points are used to initi
alize the new Gaussians (`
Gprime_box`).\n    - The re
veal is appropriate as it
follows the flow of the
\"Inpainting-guided 3DGS R
efinement\" section of the
diagram.\n\n2. **Analyze C
riterion B (Group Coherenc
e):**\n    - The elements r
evealed together are the e
dge `arrow_init` and the t
arget node `Gprime_box`.\n
    - This is a highly coheren
t group: an edge and its d
estination node. They are
```

```

structurally and logically
linked. There are no depen
dency violations, as the s
ource node (`P1_box`) was
already present.\n\n3. **C
ombining A and B:**\n    -
Criterion A is 4 (Fully Se
nsible).\n    - Criterion B
is 4 (Fully Sensible).\n
- The combined band is
4.\n\n### CRITERION_A_BAN
D: 4\n### CRITERION_B_BAN
D: 4\n### FINAL_BAND: 4\n#
## LABEL: Fully Sensible",
    "timestep": 14,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
14_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
    },
    {
        "rationale": "1. **A
nalyze Criterion A (Elemen
t Appropriateness):**\n
- Comparing the current fr
ame to the previous frame,
the new elements revealed
are:\n    - `Iprime_stack
_1`, `Iprime_stack_2`, `Ip
rime_stack_3` (the stack o
f inpainted views).\n
- `Iprime_lbl` (the label
$I'_{2:K}$).\n    - `arro
w_render_other` (the edge
from `Gprime_box` to `Ipri

```

me_stack_1')).\n - `arrow\_render\_ref` (the edge from `Gprime\_box` to `Iprime1\_img`).\n - The caption text at the bottom has also updated.\n - In the previous timestep, the refined 3DGS model (`Gprime\_box`) was already revealed. Logically, the next step is to show the output of this model: the rendered views.\n - The reveal includes both the reference view rendering path (`arrow\_render\_ref`) and the other views rendering path (`arrow\_render\_other` and the resulting `Iprime\_stack`). This is a sensible progression as it completes the "Render" phase of the "Inpainting-guided 3DGS Refinement" block.\n - Criterion A is fully satisfied.\n\n2. **Analyze Criterion B (Group Coherence):**\n\n- The elements revealed together are all related to the final output of the 3DGS model.\n- `Iprime\_stack` and `Iprime1\_img` (already present) are the targets of the two rendering arrows originating from `Gprime\_box`.\n- Th

```

ese elements are semantical
lly linked (outputs of the
same process) and structur
ally linked via the `edge`
data (`arrow_render_other`
and `arrow_render_ref`).\n
- There are no prerequisite
dependency violations; t
he source of the arrows (`
Gprime_box`) was already r
evealed.\n
- Criterion B
is fully satisfied.\n\n3.
**Combining A and B:**\n
- Both Criterion A and Cri
terion B are scored as 4.
According to the tie-break
rule, the final band is
4.",
    "criterion_a_band":
4,
    "criterion_b_band":
4,
    "final_band": 4,
    "label": "Fully Sens
ible",
    "is_valid": true,
    "raw_output": "### R
ATIONALE\n1. **Analyze Cri
terion A (Element Appropri
ateness):**\n
- Comparin
g the current frame to the
previous frame, the new el
ements revealed are:\n
- `Iprime_stack_1`, `Iprim
e_stack_2`, `Iprime_stack_
3` (the stack of inpainted

```



```

views).\n      - `Iprime_lb
l` (the label  $I'_{2:K}$ 
$).\n      - `arrow_render_
other` (the edge from `Gpr
ime_box` to `Iprime_stack_
1`).\n      - `arrow_render
_ref` (the edge from `Gpri
me_box` to `Iprime1_img
`).\n      - The caption te
xt at the bottom has also
updated.\n      - In the prev
ious timestep, the refined
3DGS model (`Gprime_box`)
was already revealed. Logi
cally, the next step is to
show the output of this mo
del: the rendered views.\n
- The reveal includes both
the reference view renderi
ng path (`arrow_render_ref
`) and the other views ren
dering path (`arrow_render
_other` and the resulting
`Iprime_stack`). This is a
sensible progression as it
completes the \"Render\" p
hase of the \"Inpainting-g
uided 3DGS Refinement\" bl
ock.\n      - Criterion A is
fully satisfied.\n\n2. **A
nalyze Criterion B (Group
Coherence):**\n      - The el
ements revealed together a
re all related to the fina
l output of the 3DGS mode
l.\n      - `Iprime_stack` an

```

```

d `Iprime1_img` (already p
resent) are the targets of
the two rendering arrows o
riginating from `Gprime_bo
x`.\n    - These elements a
re semantically linked (ou
tputs of the same process)
and structurally linked vi
a the `edge` data (`arrow_
render_other` and `arrow_r
ender_ref`).\n    - There a
re no prerequisite depende
ncy violations; the source
of the arrows (`Gprime_box
`) was already revealed.\n
- Criterion B is fully sat
isfied.\n\n3. **Combining
A and B:**\n    - Both Crit
erion A and Criterion B ar
e scored as 4. According t
o the tie-break rule, the
final band is 4.\n\n### CR
ITERION_A_BAND: 4\n### CRI
TERION_B_BAND: 4\n### FINA
L_BAND: 4\n### LABEL: Full
y Sensible",
    "timestep": 15,
    "frame_path": "/arch
ive/medha.sen/DAG_test/ben
ch/sample1_PR_TikZ/frames/
15_CVPR_2025_pipe00004_fra
mes_tikz_PR_animated.png"
  },
  {
    "rationale": "1. **A
nalyze Criterion A (Elemen

```

t Appropriateness):**\n

- Comparing the CURRENT frame to the PREVIOUS frame, the new elements revealed are the double-headed red arrow labeled $\mathcal{L}_{\text{cross}}$ (XML id: `loss\_cross`) and the corresponding caption at the bottom.\n
- In the previous timestep, the rendered views $I'_{2:K}$ (`Iprime\_stack\_1`) and the projected targets $I^P_{2:K}$ (`IP\_stack\_1`) were already present.\n
- The $\mathcal{L}_{\text{cross}}$ loss is the final logical link in the "In painting-guided 3DGS Refinement" pipeline, representing the cross-view consistency loss between these two sets of images.\n
- As this is the final timestep (16 of 16), revealing the final loss function and the concluding summary caption is the correct and logical conclusion to the animation.\n\n2. **Analyze Criterion B (Group Coherence):**\n
- The elements revealed together are the `loss\_cross` edge and the final text caption.\n
- The `loss\_cross` edge connects

cts `IP\_stack\_1` and `Ipri
me_stack_1`. Since both en
dpoints were already revea
led, the edge is logically
coherent.\n - The captio
n explicitly describes the
function of the $\mathcal{L}_{\text{cross}}$ loss ("a cro
ss-view consistency loss m
inimizes the perceptual di
fference...\").\n - Ther
e is a direct semantic and
logical link between the v
isual element (the red arr
ow) and the text explainin
g it.\n\n3. **Combining A
and B:**\n - Criterion A
is fully satisfied (4): Th
e element is the correct f
inal piece of the diagra
m.\n - Criterion B is fu
lly satisfied (4): The vis
ual element and its explan
atory text are perfectly c
oherent.\n - Final Band:
4.",

```

        "criterion_a_band":
4,
        "criterion_b_band":
4,
        "final_band": 4,
        "label": "Fully Sens
ible",
        "is_valid": true,
        "raw_output": "### R
ATIONALE\n1. **Analyze Cri

```

terion A (Element Appropriateness):**\n
 - Comparing the CURRENT frame to the PREVIOUS frame, the new elements revealed are the double-headed red arrow labeled $\mathcal{L}_{\text{cross}}$ (XML id: `loss\_cross`) and the corresponding caption at the bottom.\n
 - In the previous timestep, the rendered views $I'_{2:K}$ (`Iprime\_stack\_1`) and the projected targets $I^P_{2:K}$ (`IP\_stack\_1`) were already present.\n
 - The $\mathcal{L}_{\text{cross}}$ loss is the final logical link in the "Inpainting-guided 3DGS Refinement" pipeline, representing the cross-view consistency loss between these two sets of images.\n
 - As this is the final timestep (16 of 16), revealing the final loss function and the concluding summary caption is the correct and logical conclusion to the animation.\n\n
 Analyze Criterion B (Group Coherence):\n
 - The elements revealed together are the `loss\_cross` edge and the final text caption.\n
 - The `loss\_cross`

edge connects `IP\_stack\_1` and `Iprime\_stack\_1`. Since both endpoints were already revealed, the edge is logically coherent.

- The caption explicitly describes the function of the $\mathcal{L}_{\text{cross}}$ loss ("a cross-view consistency loss minimizes the perceptual difference...").
- There is a direct semantic and logical link between the visual element (the red arrow) and the text explaining it.

3. **Combining A and B:**

- Criterion A is fully satisfied (4): The element is the correct final piece of the diagram.
- Criterion B is fully satisfied (4): The visual element and its explanatory text are perfectly coherent.

- Final Band: 4.

CRITERION_A_BAND: 4
 #### CRITERION_B_BAND: 4
 #### FINAL_BAND: 4
 #### LABEL: Fully Sensible",
 "timestep": 16,
 "frame_path": "/archive/medha.sen/DAG_test/bench/sample1_PR_TikZ/frames/16_CVPR_2025_pipe00004_frames_tikz_PR_animated.png"

```
}  
]  
}
```